



Automated CI/CD Testing Framework for O-RAN via O2 Infrastructure Management on O-Cloud

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OUTLINE



PART 1

Introduction



PART 2

System Architecture



PART 3

Proposed Method



PART 4

Experimental Results

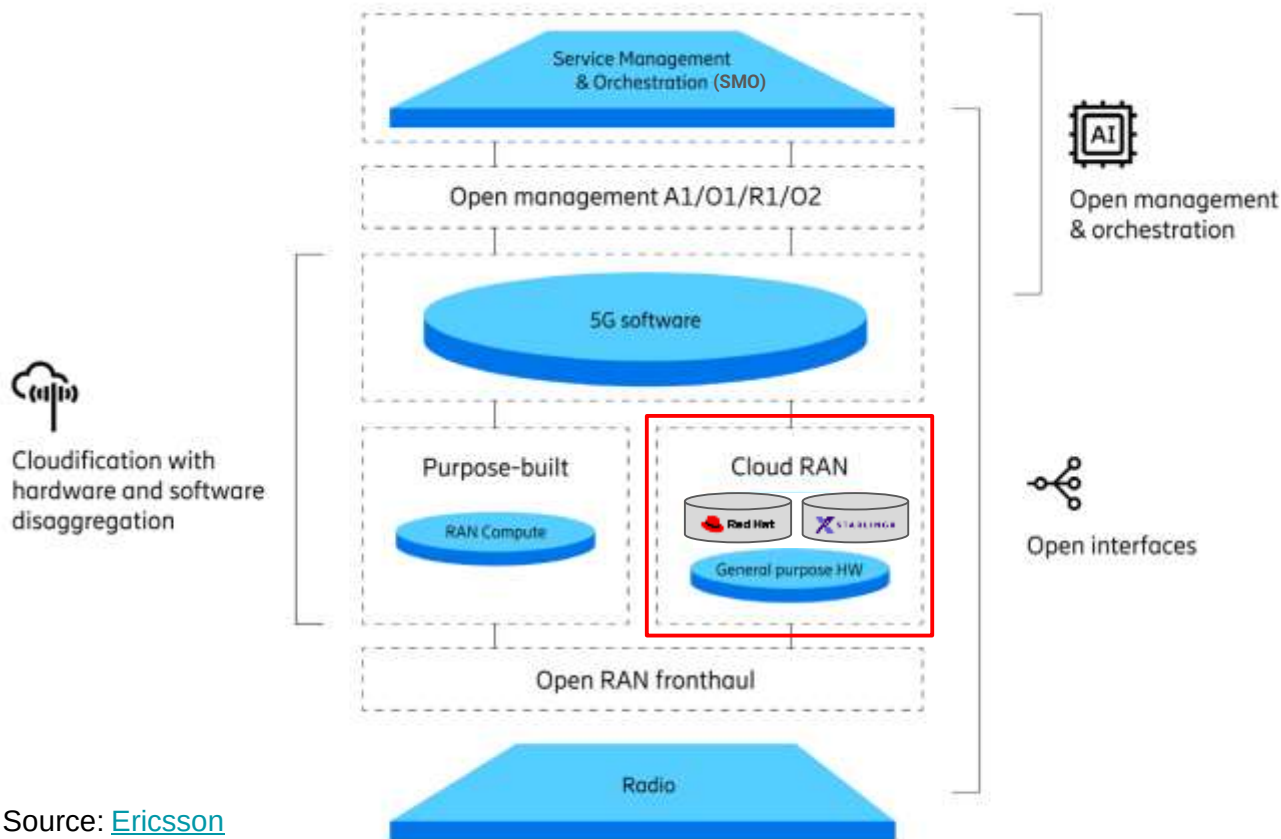


PART 5

Conclusion



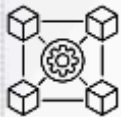
Key Elements of Open RAN (O-RAN)





Motivation

Promises of O-RAN



Openness

Promotes **interoperability** of components from different vendors [1].



Flexibility

Allows operators to **mix and match** the best hardware and software solutions [2].



Innovation

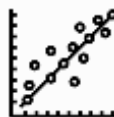
Accelerates the development cycle of new features and services [3].

Challenges of O-RAN



Integration Complexity

Components from **multi-vendors** must work together seamlessly [4].



Scale of Testing

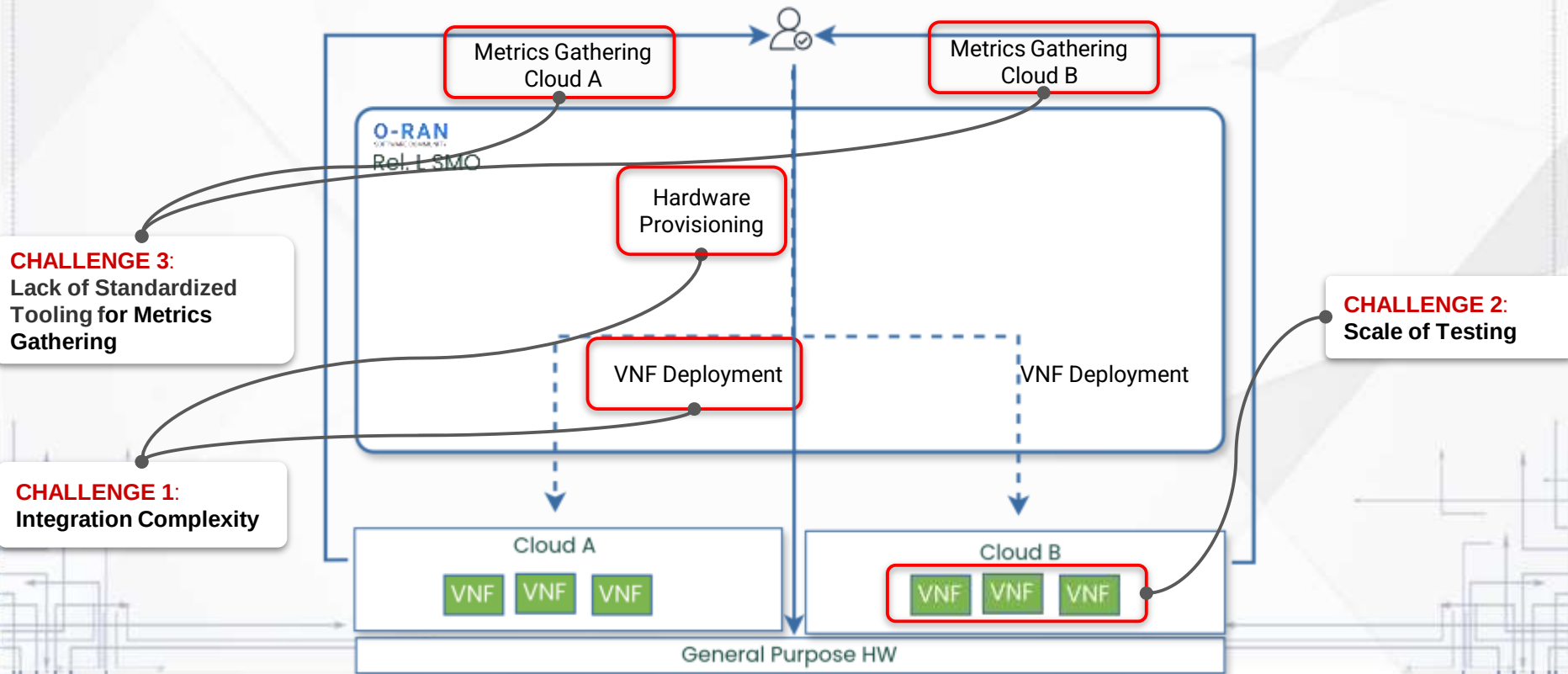
Every component update or replacement requires comprehensive **end-to-end regression testing** [5].



Lack of Standardized Tooling Different NF, O-Cloud, and O-RU combinations require separate tooling for **metrics gathering and reporting** [6].



Challenges of O-RAN





Related Work: O-Cloud

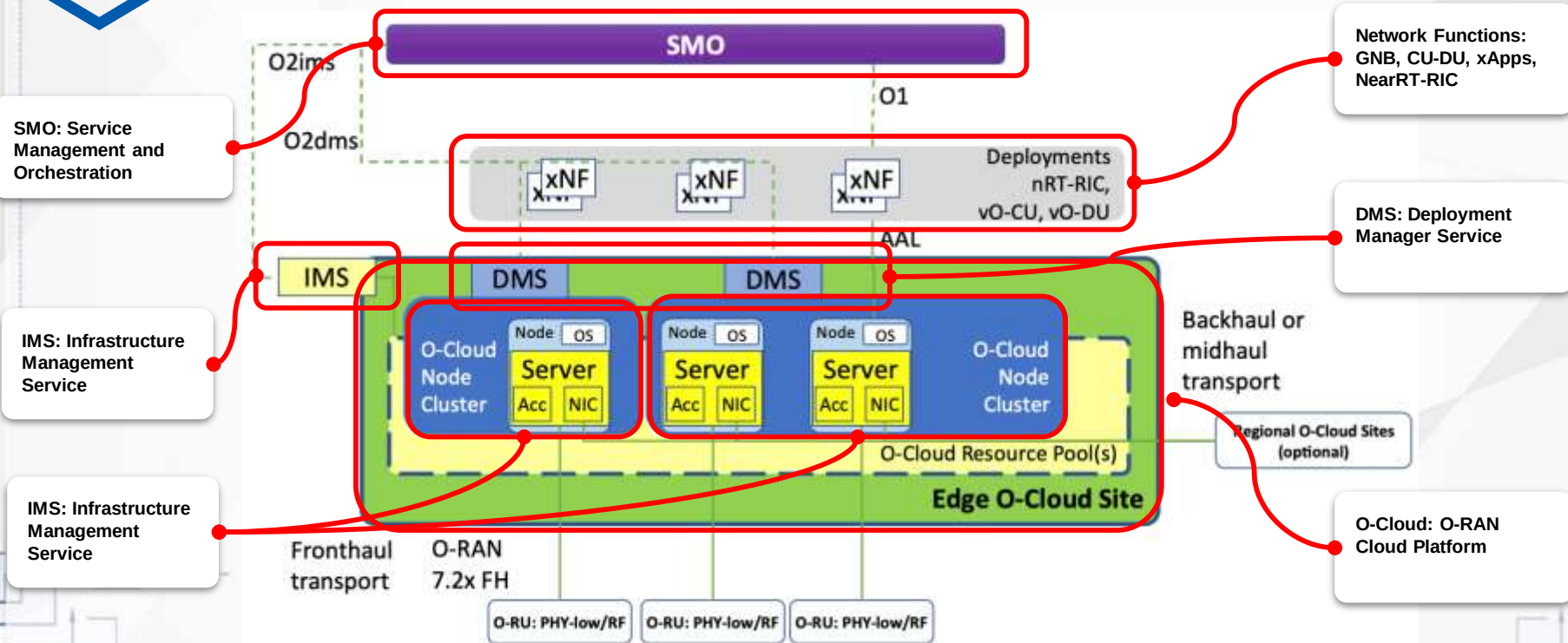


Figure 11: Edge O-Cloud Site deployment example without an O-Cloud Site Network Fabric (Single servers)



Topology & Inventory

Network Function Orchestrator
Managing deployment lifecycle events of **Network Functions**

Federated O-Cloud Orchestration and Management

Manage the distribution
of **O-Cloud** lifecycle



DME

Expose SMO **Data**
Management to
external Network



Related Works: Open Source Projects

Project	Component(s)	Role in Architecture	Notes (Standards-aligned)
StarlingX [15] [16] [17]	O-Cloud Cluster IMS DMS	Complete O-Cloud platform	Full O-Cloud realization: execution layer (nodes/clusters) and management layer (IMS/DMS). Exposes O2ims/O2dms per O-RAN WG6 v5.0
OKD [17] [18]	O-Cloud Cluster IMS DMS	Kubernetes-based O-Cloud infrastructure	Community Kubernetes distribution; alternative O-Cloud platform. Provides container execution environment with O2 operator integration
Nephio [19] [20]	SMO: FOCOM, NFO O-Cloud: IMS/DMS	GitOps automation framework	Separate instances for SMO (FOCOM/NFO) and O-Cloud (IMS/DMS automation). Intent-based lifecycle management using Kpt packages. Communicates via O2ims



Related Works: Paper

Paper	Infra Management	IMS/DMS	O-Cloud Node Cluster	O-Cloud Node	xNF	O-RU (PHY-low / RF)
Wang et al. [8]	FOCOM + AWX/Ansible	O2 IMS (custom)	K3s / KWOK (emulated)	K3s / KWOK (up to 2500 nodes)	N/A (infra focus)	N/A
Bonati et al. [9]	OpenShift GitOps	OpenShift	OpenShift (K8s)	OpenShift (K8s)	OAI gNB (CU/DU) + UE	USRP (SDR)
Liu et al. [10]	FlexRIC / OSC Near-RT RIC	N/A (bare-metal)	Ubuntu Linux (bare-metal/VM)	Ubuntu Linux (bare-metal/VM)	srsRAN gNB + srsRAN 4G (UE) Open5GS	USRP B210/X310 / ZMQ
This Work	FOCOM/NFO	Under review OSC - o-ran-sc/pti-o2 - openshift-kni/oran-o2ims	StarlingX/OKD	StarlingX/OKD	- OAI gNB Monolithic + 7.2 Mono - OAI CU/DU (F1) + 7.2 - OAI VNF/PNF (nFAPI) + 7.2 - OAI gNB RFSIM+UESim - Open5GS	LiteOn O-RU Pegatron O-RU



Motivation

Challenges

- Heterogeneous O-Cloud platforms (StarlingX & OpenShift)
- Scalability of manual testing
- Lack of standardized tooling for metrics gathering

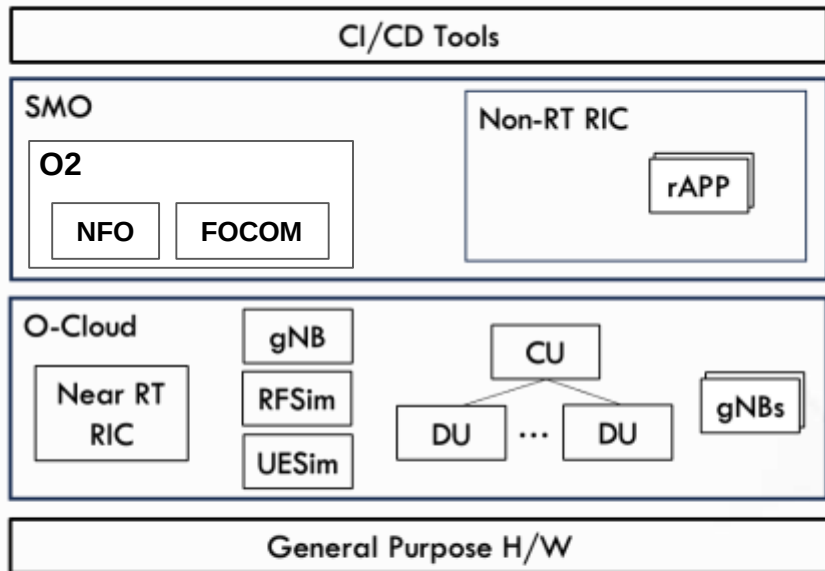
Solutions: Automated CI/CD Testing Framework for O-RAN via O2 Infrastructure Management on O-Cloud

- Platform-agnostic framework via O2 implementation of **NFO** and **FOCOM**
- CI/CD for scale of testing
- Unified profiling via sideload architecture



System Architecture

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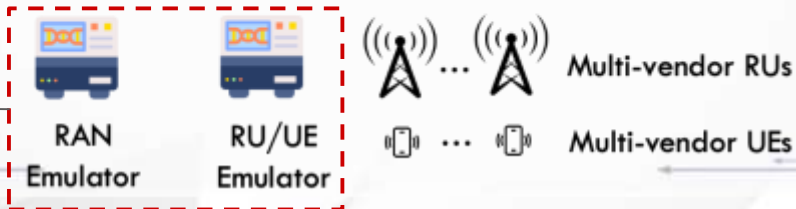


Solution 2:
CI/CD for Scale of Testing

Solution 1:
Standard-compliant Multi-vender Interoperability Test Platform

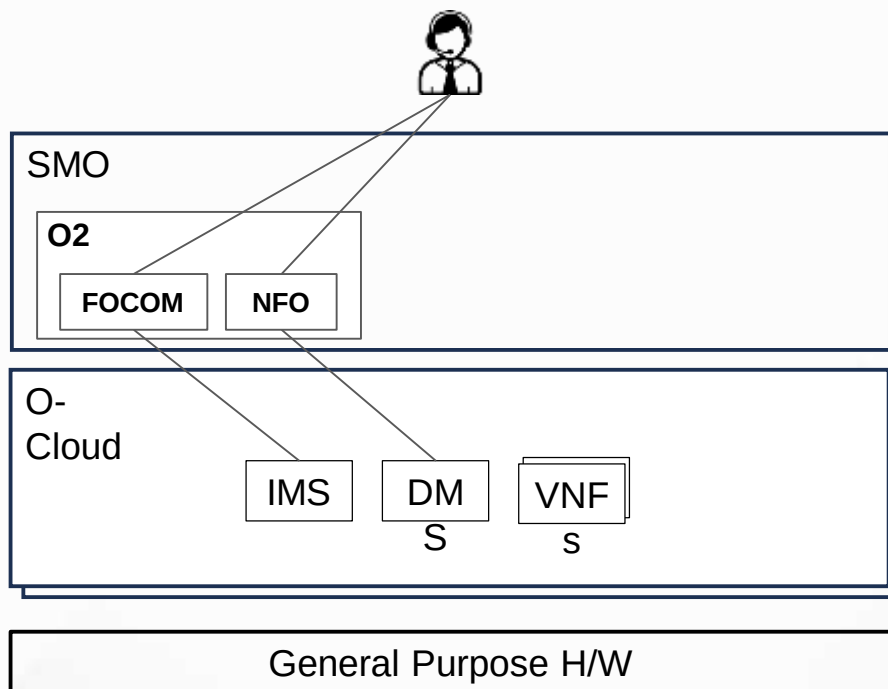
Solution 3:
Sideload Profiling

FUTURE WORK





Proposed Method: FOCOM & NFO Modules

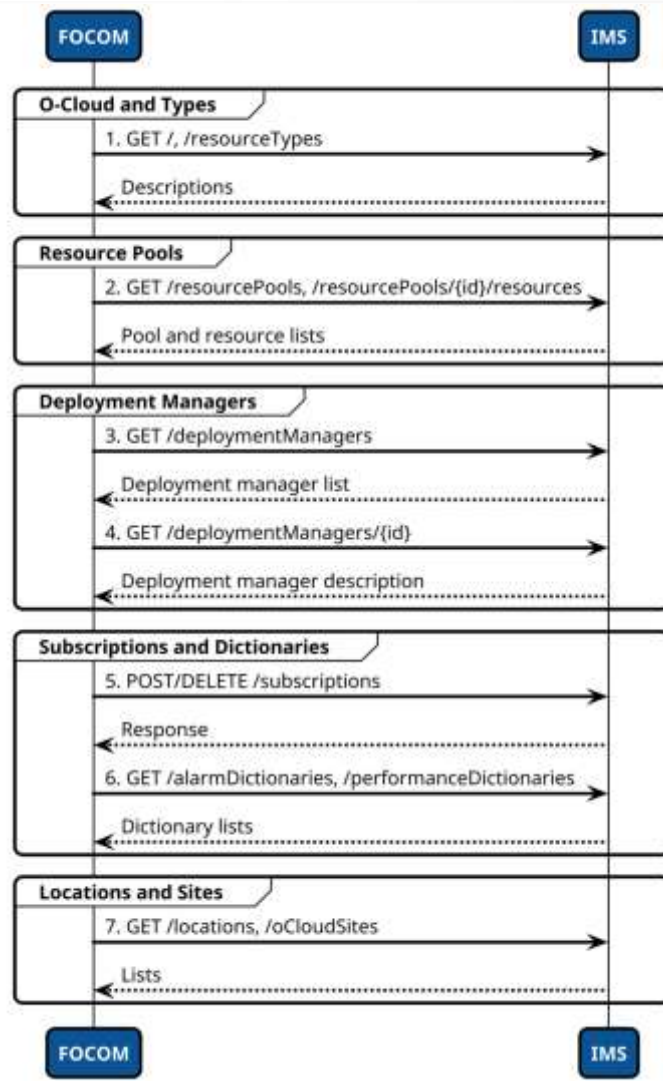




FOCOM Module

Functions:

- Multi-platform O-Cloud federation (OpenShift O2IMS, StarlingX PTI-O2)
- IMS registration and management APIs
- Deployment manager discovery and kubeconfig retrieval
- Authentication: Bearer tokens (OpenShift), OAuth2 + mTLS (PTI-O2)
- Flask REST API with SQLite backend
- Docker deployment support





FOCOM Module: Current Implementation Status

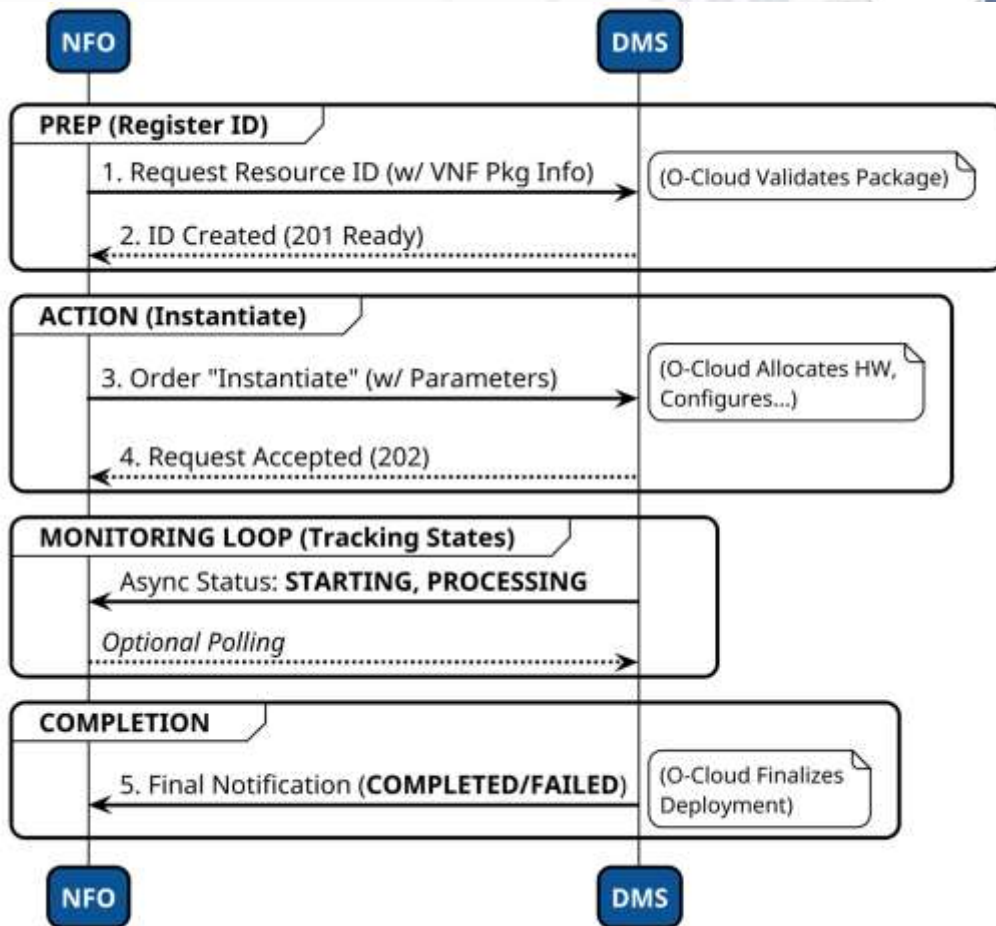
No	Resource name	Resource URI	HTTP method	OSC smo-o2	This Work
1	Resource Type List	/resourceTypes	GET	X	V
2	Resource Type Description	/resourceTypes/{resourceTypeId}	GET	X	V
3	Resource Pool List	/resourcePools	GET	X	V
4	Resource Pool Description	/resourcePools/{resourcePoolId}	GET	X	V
5	Resource List	/resourcePools/{resourcePoolId}/resources	GET	X	V
6	Resource Description	/resourcePools/{resourcePoolId}/resources/{resourceId}	GET	X	V
7	Deployment Manager List	/deploymentManagers	GET	X	V
8	Deployment Manager Description	/deploymentManagers/{deploymentManagerId}	GET	X	V



NFO Module

Function:

- Determines workload homing to appropriate O-Cloud and DMS
- Selects placement to specific cluster or location
- Deploys workload in target cluster
- Validates VNF packages against O-Cloud capabilities
- Executes Helm-based deployments
- Monitors and reports lifecycle states (STARTING, PROCESSING, COMPLETED, FAILED)



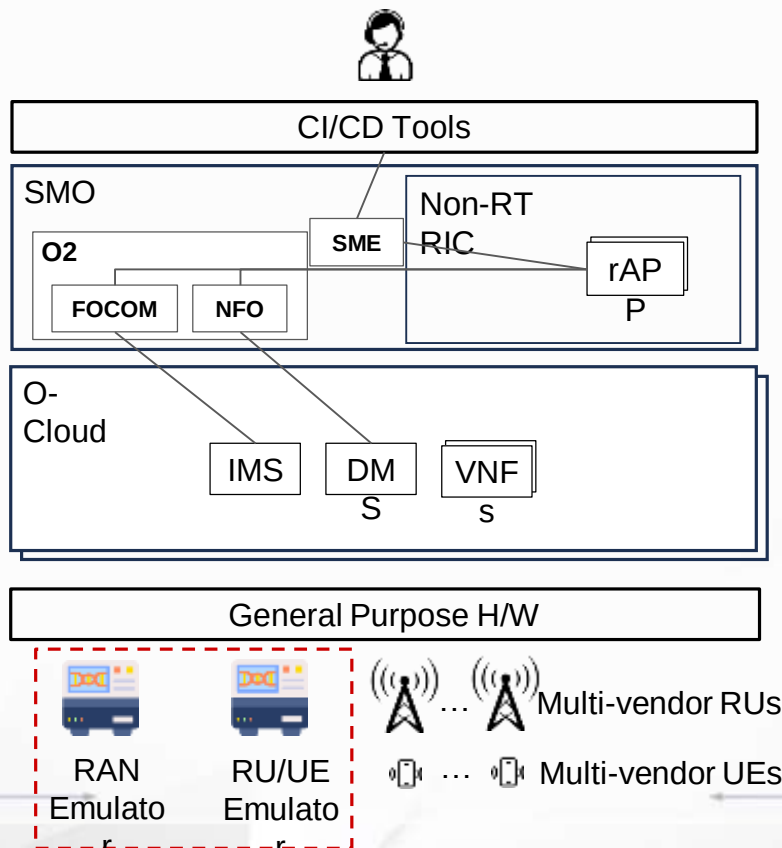


NFO Module: Current Implementation Status

No	Resource name	Resource URI	HTTP method	OSC smo-o2	This Work
1	VNF Instances	/vnf_instances	GET	X	V
			POST	X	V
2	Individual VNF instance	/vnf_instances/{vnfInstanceId}	GET	X	V
			PATCH	X	V
			DELETE	X	V
3	Instantiate VNF Task	/vnf_instances/{vnfInstanceId}/instantiate	POST	X	V
4	Terminate VNF task	/vnf_instances/{vnfInstanceId}/terminate	POST	X	V
5	Heal VNF task	/vnf_instances/{vnfInstanceId}/heal	POST	X	X
6	Scale VNF task	/vnf_instances/{vnfInstanceId}/scale	POST	X	X
7	Scale VNF to Level task	/vnf_instances/{vnfInstanceId}/scale_to_level	POST	X	X
8	Change external VNF connectivity task	/vnf_instances/{vnfInstanceId}/change_ext_connectivity	POST	X	X
...	...	...			



Proposed Method: CI/CD+rApp for automatic testing

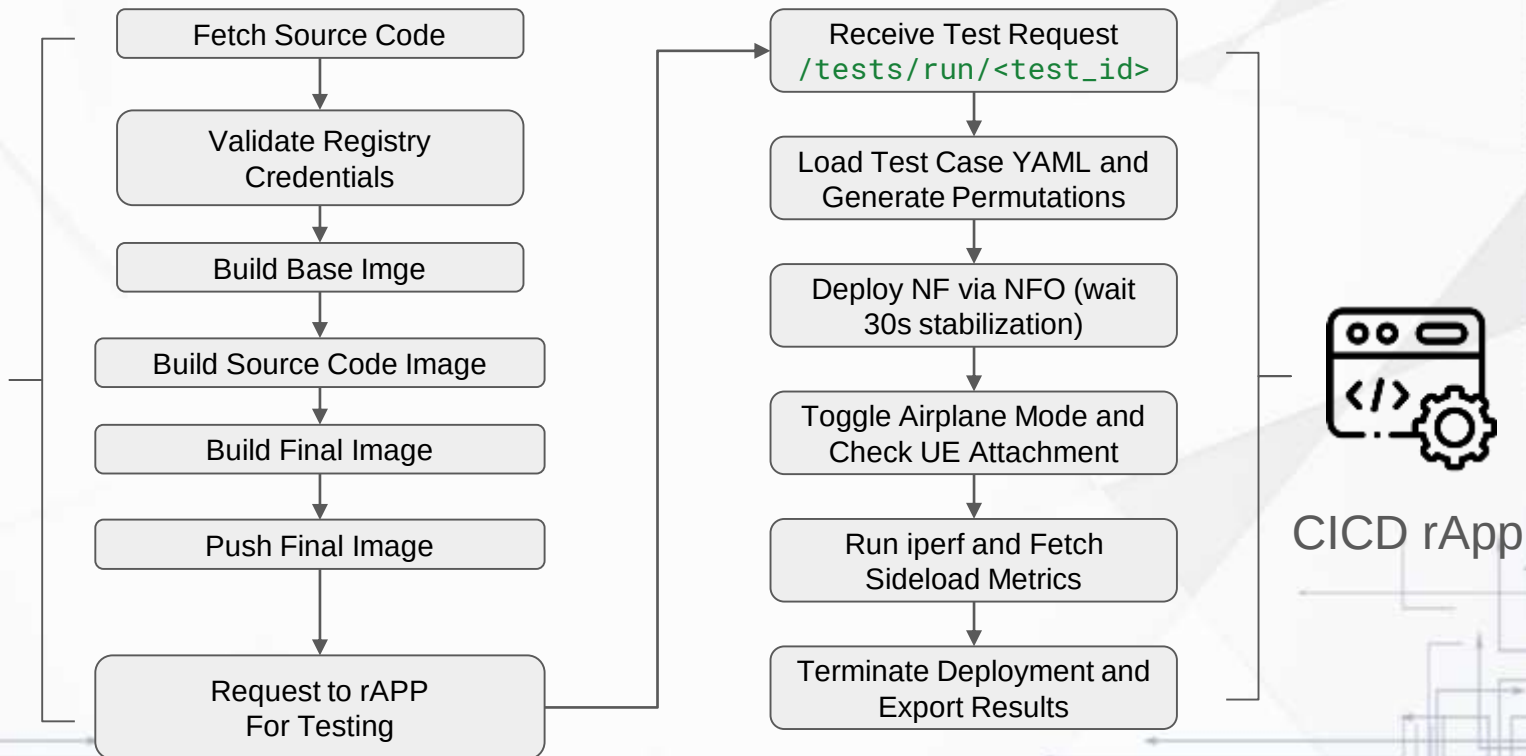




Proposed Method: CICD+rApp for automatic testing



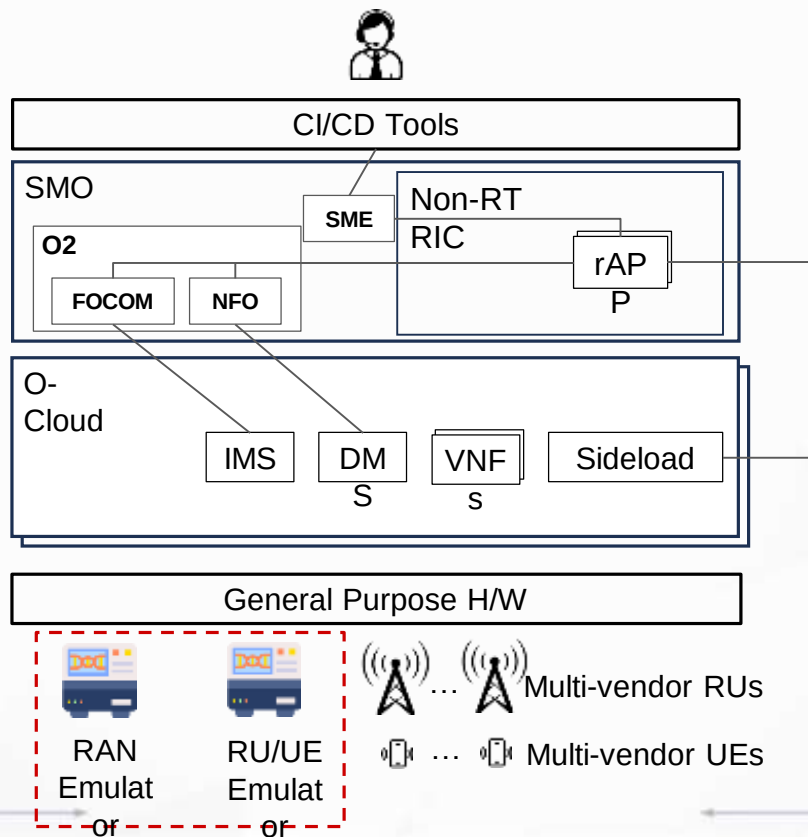
Jenkins



CICD rApp



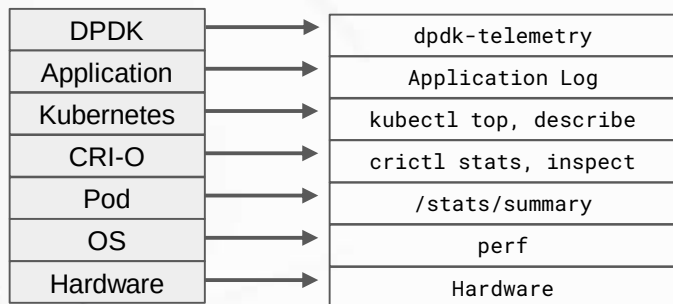
Proposed Method: Sideload Profiling



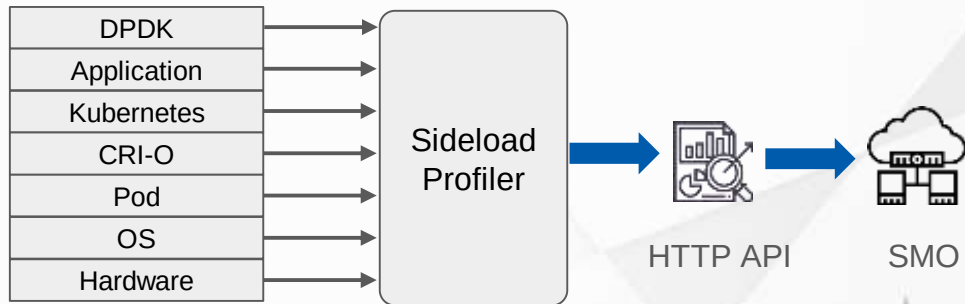


Proposed Method: Sideload Profiling

Before: Fragmented & Opaque



After: Unified & Transparent





Experiment & Validation

1. Experiment 1: [Testbed Implementation](#)
 - a. NFO and FOCOM on StarlingX and OKD O-Clouds
 - b. Different use cases
 - i. Multi-vendor interoperability test:
 1. LiteON/Pegatron
 - ii. Use Case 1: OAI GNB 7.2
 - iii. Use Case 2: OAI nFAPI
 - iv. Use Case 3: OAI F1 (CU & DU)
2. Experiment 2: Regression test of combination of VNF
 - a. GNB with different CPU allocation
 - b. GNB with different packet size
3. Experiment 3: VNF Profiling on StarlingX and OKD
 - a. CPU
 - b. Memory
 - c. Power



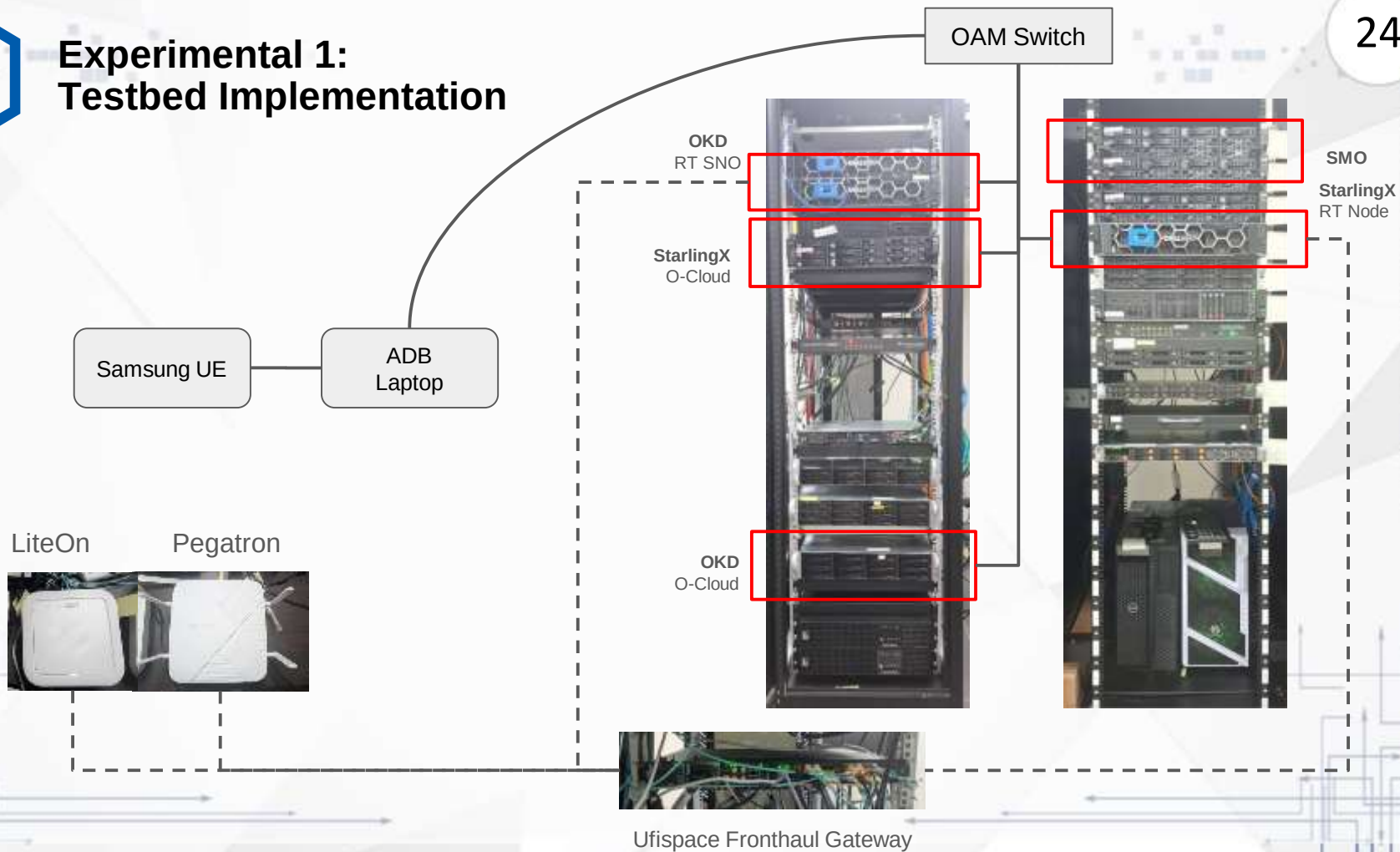
Experiment & Validation: Assumptions

1. UE already **tested**
2. Provisioning of O-Cloud is performed prior to experiment.
Due to the following reasons:
 - a. One of the O-Cloud doesn't support this yet (StarlingX).
 - b. Provisioning will be one time job. Provision the server every time test are performed is not current goal (e.g. 72 times).
3. SMO is already installed prior to this experiment.



Experimental 1: Testbed Implementation

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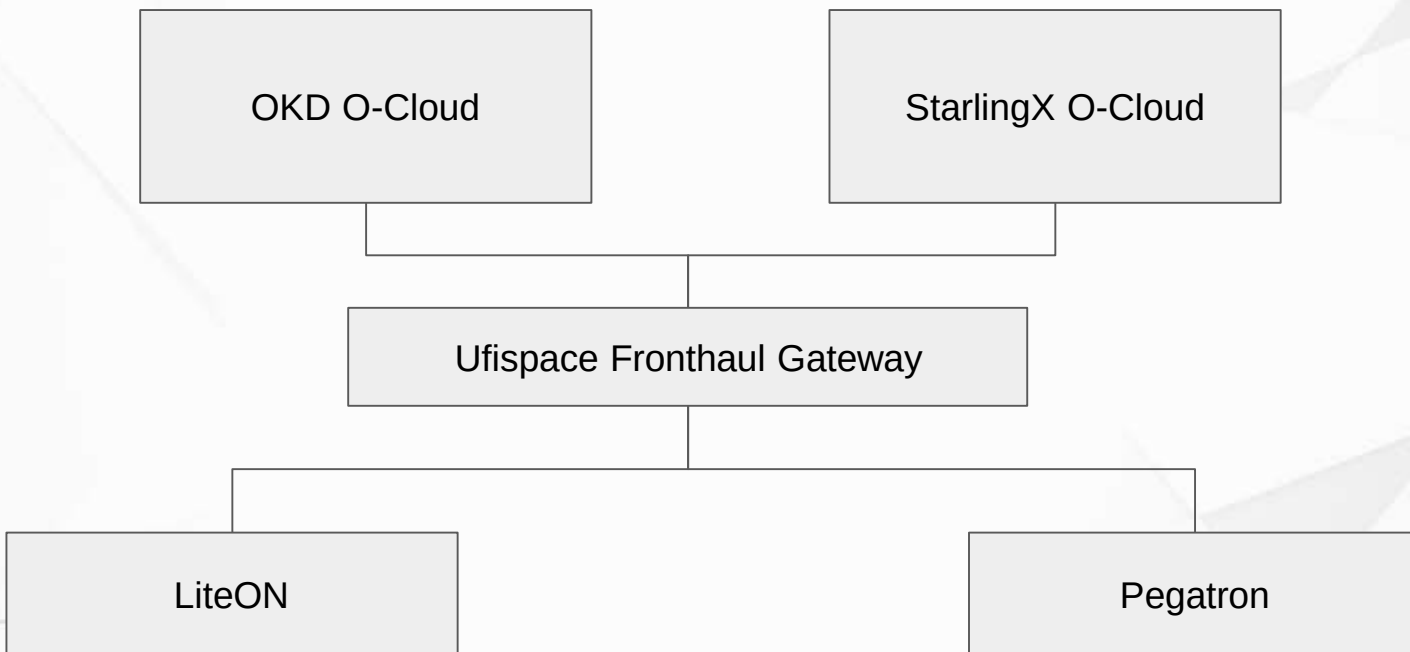
Experimental 1: Testbed Implementation

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	OKD (Kepler)	StarlingX (Joule)
CPU	Intel Xeon Gold 6326 32 cores @ 2.90GHz	Intel Xeon Gold 6326 32 cores @ 2.90GHz
RT Kernel	Stream CoreOS 9	Debian 6.6-rt
Isolated Cores	16-31 (16 cores)	Odd cores: 3,5,7...31 (16 cores)
Hugepages	40GB	40GB



Experimental 1: Integration with Different O-RU





Experimental 1: OAI GNB 7.2

```
Context: kubernetes-admin@kubernetes [R
Cluster: kubernetes
User: kubernetes-admin
K9s Rev: v0.50.16 ⚡ v0.50.18
K8s Rev: v1.29.2
CPU: n/a
Mem: n/a
Logs(oai-test/oai-gnb-metaonia-jura-77dcd5d878-ck5ph:gnb)[tail...
UE d4f4: LCID 2: TX 0 RX 0 bytes
UE d4f4: LCID 4: TX 1285608399 RX 23323 bytes
295023.314457 [HW] I [o-du 0][rx 8226796 pps 79872 kbps 7086
295023.314466 [HW] I [o-du0][pusch0 1898492 prach0 158208]
295023.314467 [HW] I [o-du0][pusch1 1898496 prach1 158208]
295023.314468 [HW] I [o-du0][pusch2 1898496 prach2 158208]
295023.314469 [HW] I [o-du0][pusch3 1898480 prach3 158208]
295024.589469 [NR_MAC] I Frame.Slot 0.0
UE RNTI d4f4 CU-UE-ID 1 in-sync PH 48 dB PCMAX 22 dBm, average
UE d4f4: CQI 15, RI 2, PMI (15,1)
UE d4f4: dlsch_rounds 78077/1905/268/75, dlsch_errors 2, pucch
UE d4f4: ulsch_rounds 4660/5/0/0, ulsch_errors 0, ulsch_0TX 1,
UE d4f4: MAC: TX 1300261207 RX 434237 bytes
UE d4f4: LCID 1: TX 845 RX 3095 bytes
UE d4f4: LCID 2: TX 0 RX 0 bytes
UE d4f4: LCID 4: TX 1285608399 RX 23323 bytes
295024.594451 [HW] I [o-du 0][rx 8306668 pps 79872 kbps 3543
295024.594460 [HW] I [o-du0][pusch0 1916924 prach0 159744]
295024.594461 [HW] I [o-du0][pusch1 1916928 prach1 159744]
295024.594462 [HW] I [o-du0][pusch2 1916928 prach2 159744]
295024.594473 [HW] I [o-du0][pusch3 1916912 prach3 159744]
<pod> <logs>
```

Kubernetes

GNB Software



Experimental 1: OAI nFAPI

PNF

```
root@oai-pnf-pegatron-5f64ddc64-qg9p: /opt/oai-grb

Context: kubernet-es-admin@kubernet-es [94]
Cluster: kubernet-es
User: kubernet-es-admin
K8s Rev: v0.50.16
K8s Rev: v1.29.2
CPU: n/a
MEM: n/a

Logs(oai-pnf/oai-pnf-pegatron-5f64ddc64-qg9p) (tail)
Autoscroll: [x] Fullscreen: [x] Timescale: [x] Graph: [x]
857573.752868 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.754847 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.757418 [W] 50321088: check_nr_p7_early_at_execution: dl_tti_req-
857573.757424 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.757843 [W] 50321088: check_nr_p7_early_at_execution: dl_tti_req-
857573.759966 [W] 50321088: check_nr_p7_early_at_execution: dl_tti_req-
857573.759974 [W] 50321088: check_nr_p7_early_at_execution: tx_data_req-
857573.759976 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.760384 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.762411 [W] 50321088: check_nr_p7_early_at_execution: dl_tti_req-
857573.762418 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.767483 [W] 50321088: check_nr_p7_early_at_execution: dl_tti_req-
857573.767490 [W] 50321088: check_nr_p7_early_at_execution: tx_data_req-
857573.767492 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.767931 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.772361 [W] 50321088: check_nr_p7_early_at_execution: dl_tti_req-
857573.772368 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.774865 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.777419 [W] 50321088: check_nr_p7_early_at_execution: dl_tti_req-
857573.777426 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.777853 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.782415 [W] 50321088: check_nr_p7_early_at_execution: dl_tti_req-
857573.782422 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-
857573.787490 [W] 50321088: check_nr_p7_early_at_execution: dl_tti_req-
857573.787497 [W] 50321088: check_nr_p7_early_at_execution: tx_data_req-
857573.787499 [W] 50321088: check_nr_p7_early_at_execution: ul_tti_req-

[ctrl-c] [logs]
```

VNF

```
root@oai-vnf-798497545f-4c25: /opt/oai-grb

Context: kubernet-es-admin@kubernet-es [94]
Cluster: kubernet-es
User: kubernet-es-admin
K8s Rev: v0.50.16
K8s Rev: v1.29.2
CPU: n/a
MEM: n/a

Logs(oai-test/oai-vnf-798497545f-4c25) (tail)
Autoscroll: [x] Fullscreen: [x] Timescale: [x] Graph: [x]
UE dfec: disch_rounds 95953/1217/122/13, disch_errors 0, pucch0_DTX 0, BLER 0.04126 M
UE dfec: ulsch_rounds 8956/34/0/0, ulsch_errors 0, ulsch_DTX 0, BLER 0.00084 MCS (0)
UE dfec: MAC: TX 2777392454 RX 867790 bytes
UE dfec: LCID 1: TX 1365 RX 6867 bytes
UE dfec: LCID 2: TX 0 RX 0 bytes
UE dfec: LCID 4: TX 2768316598 RX 87775 bytes

857580.621130 [NR_MAC] I [1766558444.713] Frame.Slot 384.0
UE INTI dfec CU-UE-ID 1 in-sync PH 44 dB PCMAX 21 dBm, average RSRP -82 (32 meas)
UE dfec: CQI 15, RI 2, PMI (15,1)
UE dfec: disch_rounds 97067/1243/124/13, disch_errors 0, pucch0_DTX 0, BLER 0.02404 M
UE dfec: ulsch_rounds 9076/34/0/0, ulsch_errors 0, ulsch_DTX 0, BLER 0.00024 MCS (0)
UE dfec: MAC: TX 2892705484 RX 880616 bytes
UE dfec: LCID 1: TX 1368 RX 6829 bytes
UE dfec: LCID 2: TX 0 RX 0 bytes
UE dfec: LCID 4: TX 2875285933 RX 88158 bytes

857581.901129 [NR_MAC] I [1766558445.993] Frame.Slot 512.0
UE INTI dfec CU-UE-ID 1 in-sync PH 44 dB PCMAX 21 dBm, average RSRP -82 (32 meas)
UE dfec: CQI 15, RI 2, PMI (15,1)
UE dfec: disch_rounds 99792/1257/127/13, disch_errors 0, pucch0_DTX 0, BLER 0.01200 M
UE dfec: ulsch_rounds 9194/34/0/0, ulsch_errors 0, ulsch_DTX 0, BLER 0.00007 MCS (0)
UE dfec: MAC: TX 3804144014 RX 893124 bytes
UE dfec: LCID 1: TX 1374 RX 6873 bytes
UE dfec: LCID 2: TX 0 RX 0 bytes
UE dfec: LCID 4: TX 2966280687 RX 88287 bytes

[ctrl-c] [logs]
```




Experimental 1: OAI F1 (CU & DU)

CU

DU

```
Context: admin
Cluster: kepler-du
User: admin
K9s Rev: v0.32.5 v0.50.18
K8s Rev: v1.32.8-dirty
CPU: 2%
MEM: 9%
```

```
Logs(default/oai-cu-5668b4f486-fqblp:oai-cu)[tail]
Autoscroll:On FullScreen:Off Timestamps:Off Wrap:Off
332932.909793 [GTPU] I Initializing UDP for local address 10.20.9.99 with
332932.909827 [GTPU] I Created gtpu instance id: 95
332932.910306 [UTIL] I threadCreate() for time source realtime: creating t
332932.910355 [UTIL] I time manager configuration: [time source: realtime]
332932.913621 [NGAP] A Send NGSetupRequest to AMF
332932.913629 [NGAP] I 3584 -> 0000e000
332932.914817 [NGAP] I Supported PLMN 0: MCC=001 MNC=01
332932.914821 [NGAP] I Supported slice (PLMN 0): SST=0x01 SD=000
332932.914823 [NGAP] I Supported slice (PLMN 0): SST=0x01 SD=000
332932.914825 [NGAP] I Supported slice (PLMN 0): SST=0x01 SD=123
332932.914826 [NGAP] I Supported slice (PLMN 0): SST=0x01 SD=173451
332932.914827 [NGAP] A Received NGSetupResponse from AMF
332932.914838 [GNB_APP] I [gnb 0] Received NGAP_REGISTER_GNB_CNF: associat
332938.690870 [NR_RRC] I Received F1 Setup Request from gNB_DU 3605 (oai-d
332938.690942 [NR_RRC] I Accepting DU 3605 (oai-du), sending F1 Setup Resp
332938.690945 [NR_RRC] I DU uses RRC version 17.3.0
332938.691167 [NR_RRC] I cell PLMN 001.01 Cell ID 1 is in service
```

<pod> <logs>

```
Context: admin
Cluster: kepler-du
User: admin
K9s Rev: v0.32.5 v0.50.18
K8s Rev: v1.32.8-dirty
CPU: 2%
MEM: 9%
```

```
Logs(default/du-7895546974-2ncsv)[tail]
Autoscroll:On FullScreen:Off Timestamps:Off Wrap:Off
du 332942.917859 [PHY] I got sync (L1_stats_thread)
du 332945.655633 [HW] I first_call set from phy cb
du 332945.655999 [HW] I before adjusting, OAI: frame=0 slot=0, XTRAN: frame=0
du 332945.656010 [HW] I After adjusting, OAI: frame=0 slot=9, XTRAN: frame=0
du 332946.931359 [HW] I [o-du 0][rx 39936 pps 39936 kbps 1487232][tx 12
du 332946.931374 [HW] I [o-du0][pusch0 9216 prach0 768]
du 332946.931376 [HW] I [o-du0][pusch1 9216 prach1 768]
du 332946.931378 [HW] I [o-du0][pusch2 9216 prach2 768]
du 332946.931380 [HW] I [o-du0][pusch3 9216 prach3 768]
du 332947.665455 [PHY] A Command line parameters for OAI UE: -C 3750000000 -
du 332948.205389 [NR_MAC] I Frame.Slot 256.0
du
du 332948.211358 [HW] I [o-du 0][rx 79872 pps 39936 kbps 1771622][tx 25
du 332948.211374 [HW] I [o-du0][pusch0 18432 prach0 1536]
du 332948.211376 [HW] I [o-du0][pusch1 18432 prach1 1536]
du 332948.211382 [HW] I [o-du0][pusch2 18432 prach2 1536]
du 332948.211384 [HW] I [o-du0][pusch3 18432 prach3 1536]
```

<pod> <logs>



Experiment 2: Experimental Settings

O-Cloud	Split	O-RU	CPU	Traffic
StarlingX	[MONO, F1, NFAPI]	[Liteon, Pegatron]	[8, 14]	[100, 400, 700]
OKD	[MONO, F1, NFAPI]	[Liteon, Pegatron]	[8, 14]	[100, 400, 700]

Total Tests: **72 Scenario**



Experiment 2: Experimental Settings Automation Template

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Test Template on rApp

TEIV Notation of GNB and O-RU under test

O-Cloud specific, where to deploy and test

Deployment Intent source and details

Deployment Intent source and details

```
testId: pegatron-joule-cpu-bandwidth
testType: BANDWIDTH_SCALING
description: "Bandwidth baseline test with 100Mbps iperf load - Pegatron O-RU - Max CPU"
target:
  oduUrn: urn:o-ran:gnb:joule-pegatron
  cellUrn: urn:o-ran:cell:joule-pegatron
  sideloadInstanceId: 139d2f51-b3c2-4a94-a6a3-401464537f23
  cluster: ccf397ba-b1c4-4a3e-bc8d-6af58ef53818
  branch: starlingx/pegatron
  artifactRepoUrl: https://github.com/motangpuar/ocloud-helm-templates.git
  artifactName: oai-gnb-fhi-72
execution:
  runsPerCase: 1
  stabilizationTime: 30
  iperfDuration: 60
parameters:
  wr_isolcpus: [8,14]
  iperf_bandwidth_mbps: [100,400,700]
baseline:
  wr_isolcpus: 10
  memory: 16Gi
  hugepages: 10Gi
extra:
  amfhost: 192.168.8.82
```



Experiment 2: Experimental Settings Automated Pipeline

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Trigger Bandwidth Tests - 24m 37s

```
o echo "List Available Tet" curl -X POST http://192.168.8.35:5000/tests/lists curl -X POST http://192.168.8.35:5000/tests/run/liteon-joule-cpu-bandwidth curl -X POST... — Shell Script 24m 36s
```

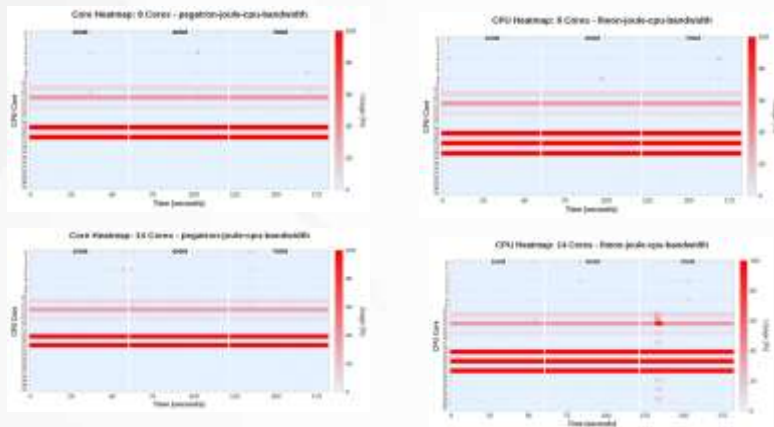
```
1 + echo "List Available Tet"
2 list Available Tet
3 + curl -X POST http://192.168.8.35:5000/tests/lists
4   % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
5   % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
6   0     0    0     0     0     0      0      0  --:--:-- --:--:-- --:--:--    0    0     0     0     0     0  0.00  0.00     0     0
7 00000
8 <!doctype html>
9 <html lang=en>
10 <title>404 Not Found</title>
11 <h1>Not Found</h1>
12 <p>The requested URL was not found on the server. If you entered the URL manually please check your spelling and try again.</p>
13 + curl -X POST http://192.168.8.35:5000/tests/run/liteon-joule-cpu-bandwidth
14   % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
15   % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
```



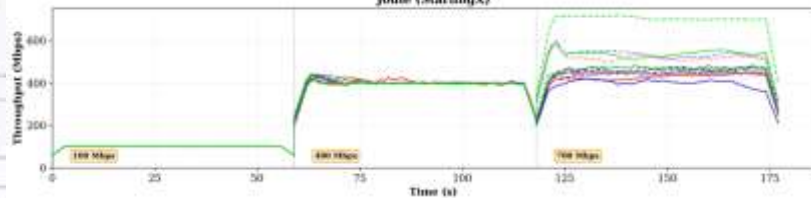

Experimental 2

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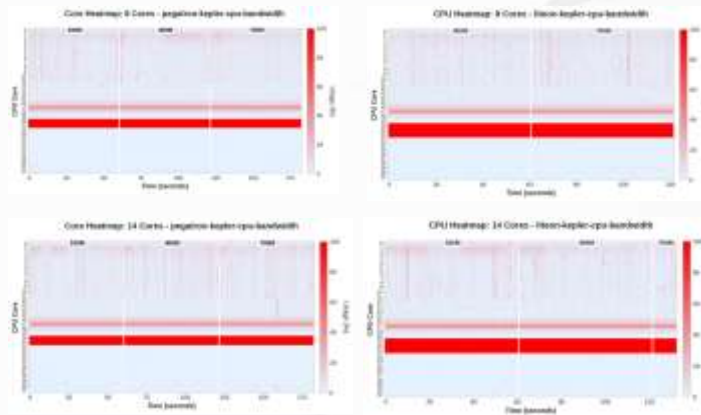
StarlingX



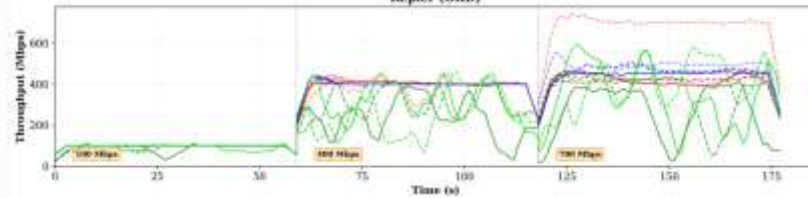
Joule (StarlingX)



OKD



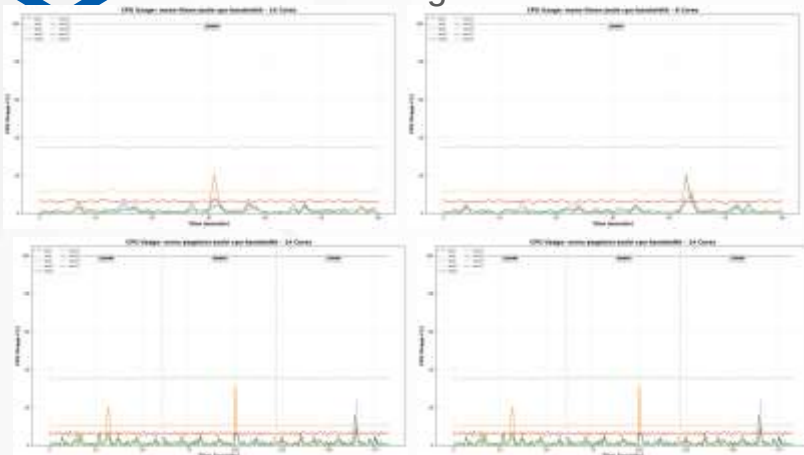
Kepler (OKD)



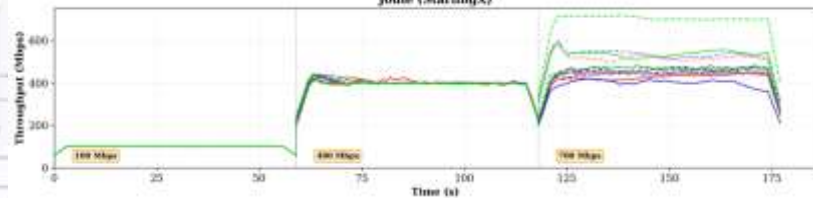


Experimental 2

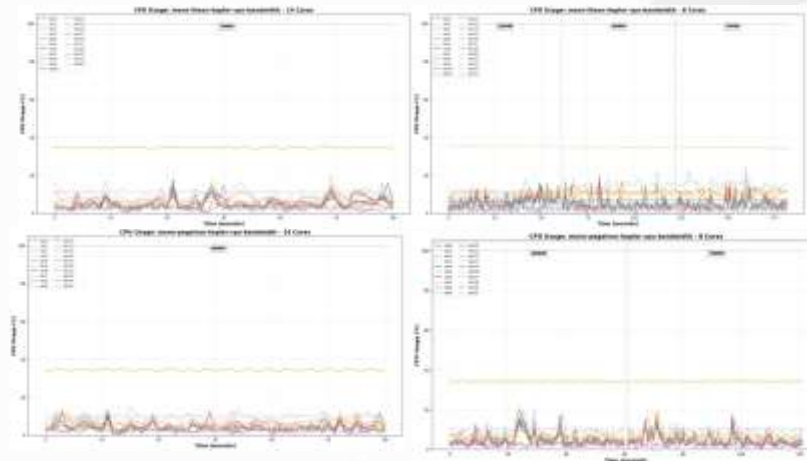
StarlingX



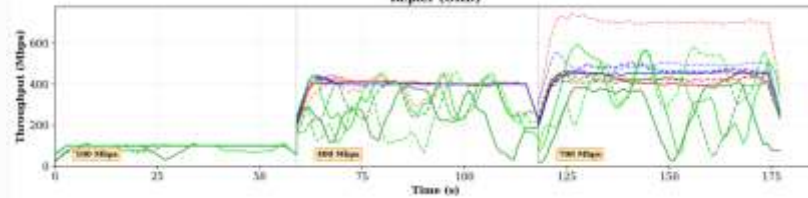
Joule (StarlingX)



OKD



Kepler (OKD)

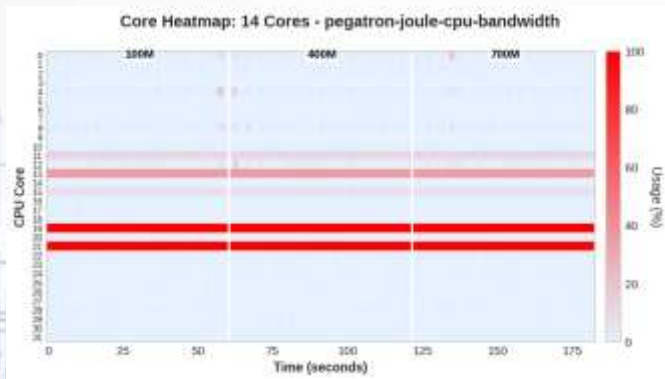
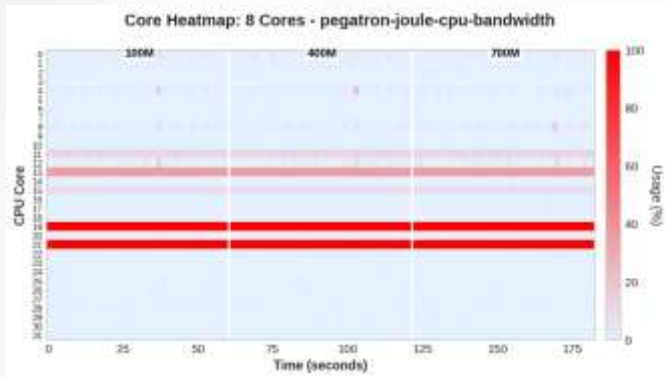




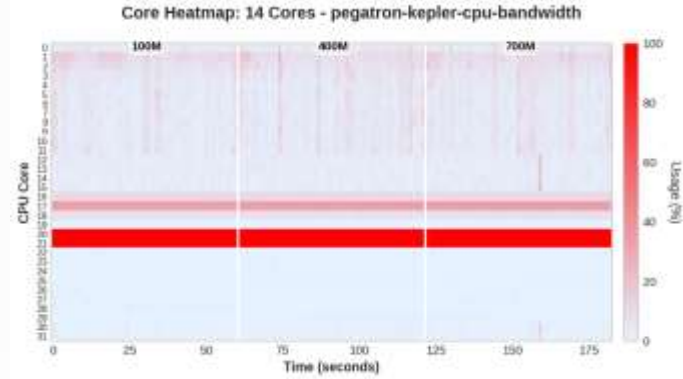
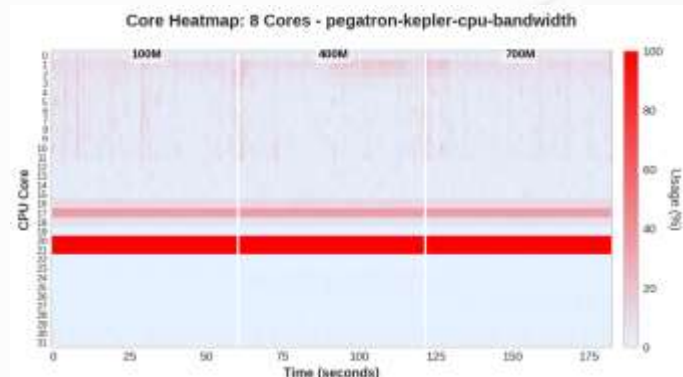
Experimental 2: GNB with different CPU allocation (Pegatron)

35

StarlingX



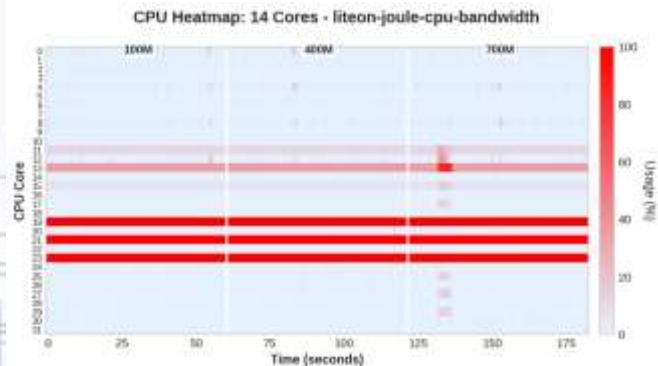
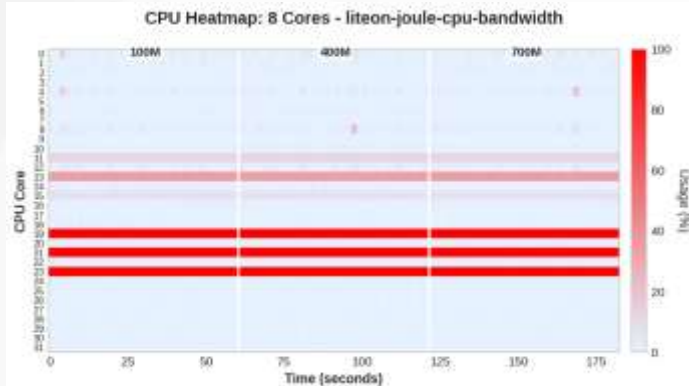
OKD



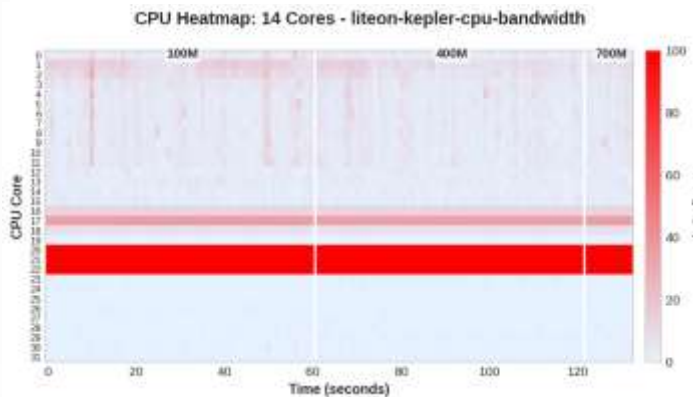
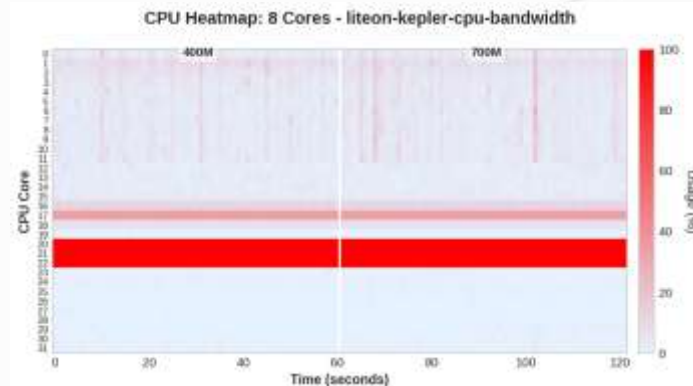


Experimental 2: GNB with different CPU allocation (LiteOn) 36

StarlingX



OKD

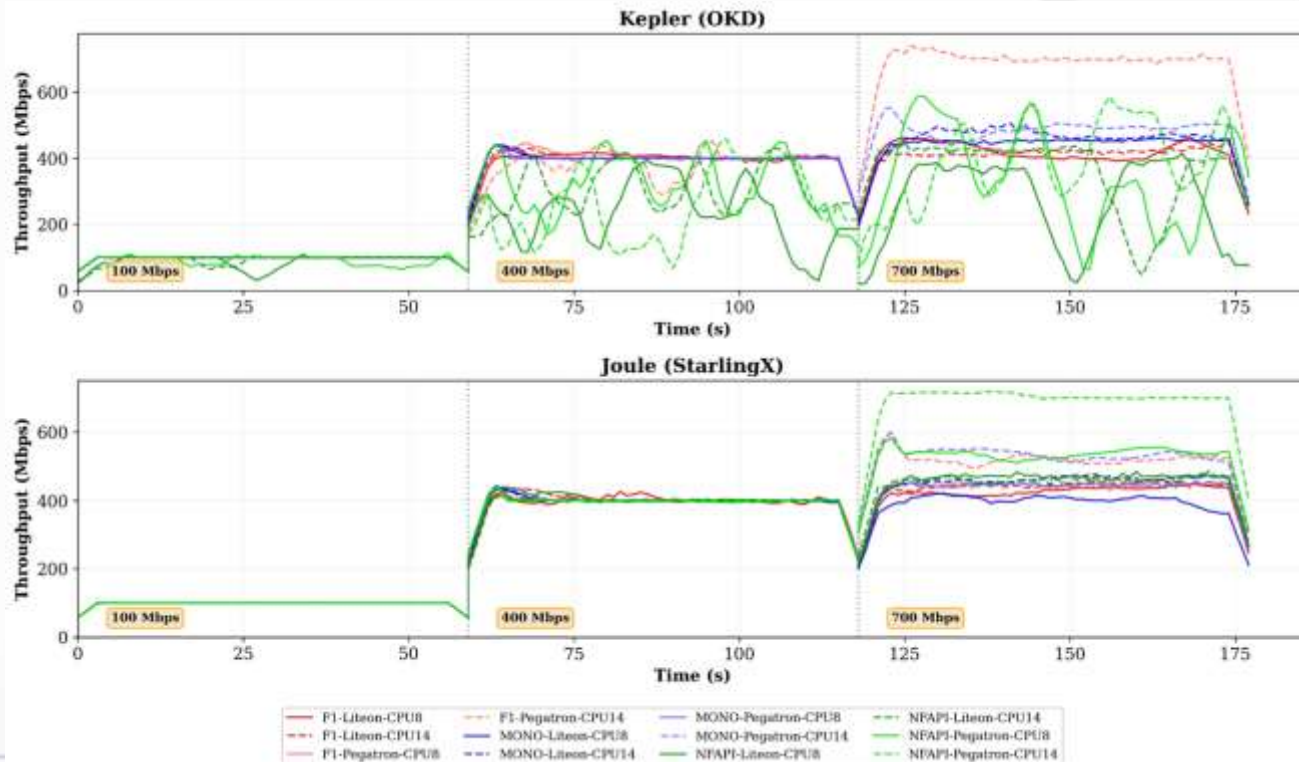




Experimental 2: NF with different CPU allocation StarlingX & OKD O-Cloud

37

- StarlingX maintains stable throughput across all load levels with minimal variance
- OKD exhibits significant throughput instability at 700 Mbps, particularly in 8-core configurations
- StarlingX achieves higher peak throughput with better consistency under maximum load
- OKD's throughput variance indicates resource contention or scheduling inefficiencies at high bandwidth



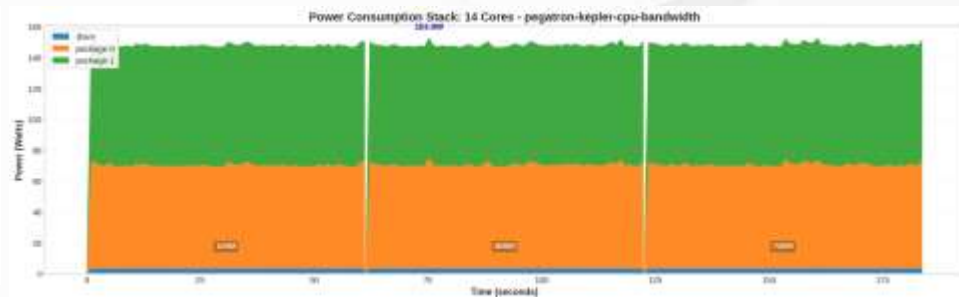
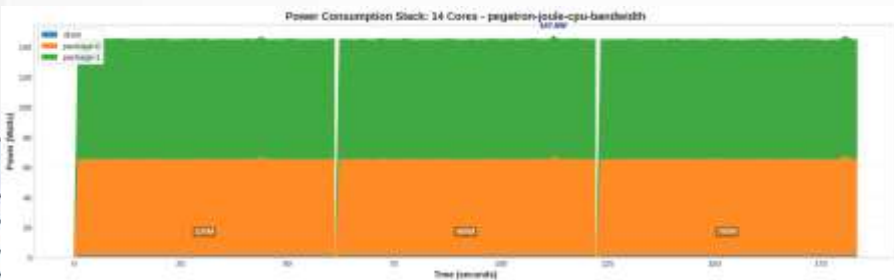
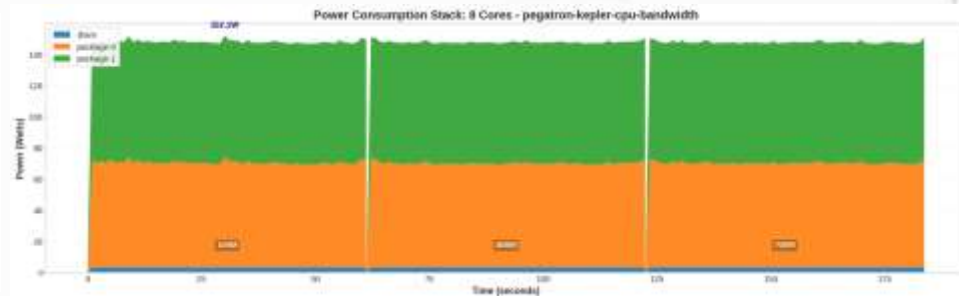
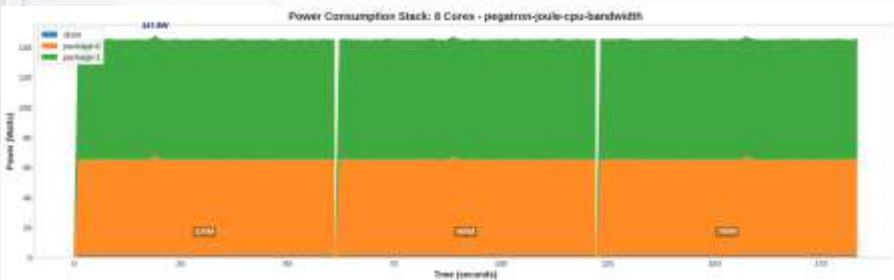


Experimental 2: GNB with different CPU allocation (Pegatron)

38

StarlingX

OKD

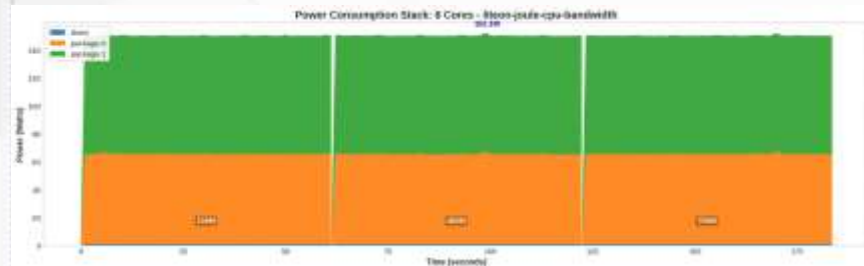




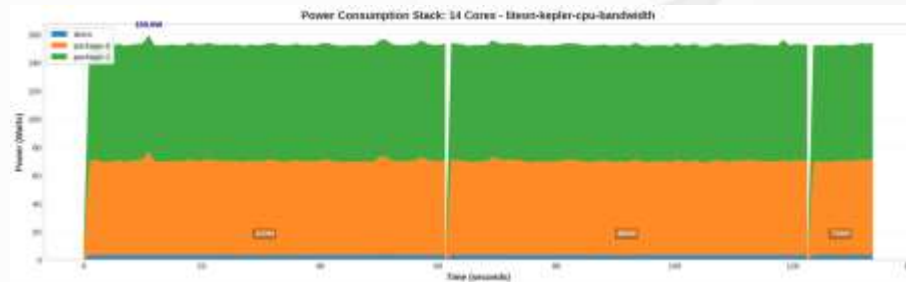
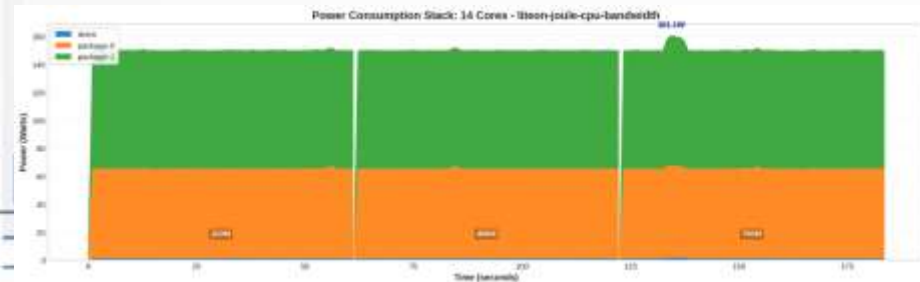
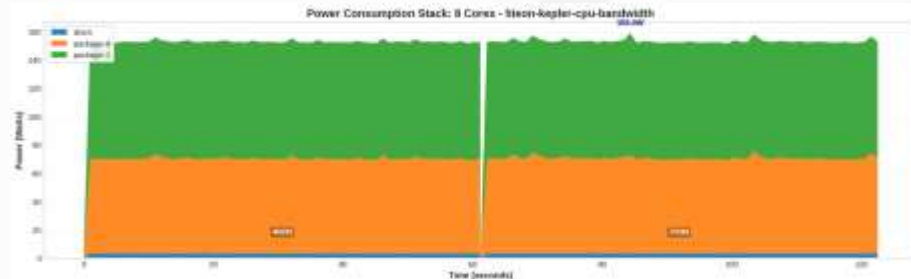
Experimental 2: GNB with different CPU allocation (LiteOn)

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StarlingX



OKD





Experimental 2: Regression Test Summary

- LiteON passes all tests on both platforms
- Pegatron fails at 8-core/700Mbps on OKD but passes on StarlingX
- Scaling to 14 cores resolves all Pegatron failures
- OKD requires higher CPU allocation than StarlingX for equivalent high-bandwidth workloads

Brand	Split	Cores	OKD Stability			StarlingX Stability		
			100	400	700	100	400	700
LiteON	F1	8c	PASS	PASS	PASS	PASS	PASS	PASS
LiteON	F1	14c	PASS	PASS	PASS	PASS	PASS	PASS
LiteON	MONO	8c	PASS	PASS	PASS	PASS	PASS	PASS
LiteON	MONO	14c	PASS	PASS	PASS	PASS	PASS	PASS
LiteON	NEAPI	8c	PASS	PASS	PASS	PASS	PASS	PASS
LiteON	NEAPI	14c	PASS	PASS	PASS	PASS	PASS	PASS
Pegatron	F1	8c	PASS	PASS	FAILED	PASS	PASS	PASS
Pegatron	F1	14c	PASS	PASS	PASS	PASS	PASS	PASS
Pegatron	MONO	8c	PASS	PASS	FAILED	PASS	PASS	PASS
Pegatron	MONO	14c	PASS	PASS	PASS	PASS	PASS	PASS
Pegatron	NEAPI	8c	PASS	PASS	PASS	PASS	PASS	PASS
Pegatron	NEAPI	14c	PASS	PASS	PASS	PASS	PASS	PASS



Experimental 2: GNB Error Log Dump

GNB crash log monolithic
mode with 700Mbps
Traffic with 8 CPU
allocation

```
17684.005657 [HW] E Detected double sync message 671.4 => 671.5
17684.005666 [HW] E Received Time doesn't correspond to the time we think it is (slot mismatch, received 671.5, expected 671.4)
17684.007580 [NR_MAC] W UE 55b2 expected HARQ pid 1 feedback at 670.19, but is at 671.3 instead (HARQ feedback is in the past)
17684.009259 [NR_MAC] E current PUCCH inactive: curr_pucch frame.slot 671.9 not matching with computed frame.slot 671.19
17684.012370 [NR_MAC] W UE 55b2 expected HARQ pid 10 feedback at 671.4, but is at 671.13 instead (HARQ feedback is in the past)
17684.012380 [NR_MAC] W UE 55b2 expected HARQ pid 7 feedback at 671.4, but is at 671.13 instead (HARQ feedback is in the past)
17684.012382 [NR_MAC] W UE 55b2 expected HARQ pid 13 feedback at 671.9, but is at 671.13 instead (HARQ feedback is in the past)
17684.012384 [NR_MAC] W UE 55b2 expected HARQ pid 6 feedback at 671.9, but is at 671.13 instead (HARQ feedback is in the past)
17684.070738 [HW] E Detected double sync message 677.14 => 677.15
17684.070767 [HW] E Received Time doesn't correspond to the time we think it is (slot mismatch, received 677.15, expected 677.14)
17684.072611 [HW] W we get rach data from ru, but it is not scheduled 677.19
17684.075083 [NR_MAC] W UE 55b2 expected HARQ pid 0 feedback at 677.14, but is at 677.18 instead (HARQ feedback is in the past)
17684.075097 [NR_MAC] W UE 55b2 expected HARQ pid 3 feedback at 677.14, but is at 677.18 instead (HARQ feedback is in the past)
17684.075744 [HW] E Detected double sync message 678.4 => 678.5
17684.075755 [HW] E Received Time doesn't correspond to the time we think it is (slot mismatch, received 678.5, expected 678.4)
17684.076172 [NR_MAC] E current PUCCH inactive: curr_pucch frame.slot 677.19 not matching with computed frame.slot 678.9
17684.077683 [NR_MAC] W UE 55b2 expected HARQ pid 2 feedback at 677.19, but is at 678.3 instead (HARQ feedback is in the past)
17684.077694 [NR_MAC] W UE 55b2 expected HARQ pid 1 feedback at 677.19, but is at 678.3 instead (HARQ feedback is in the past)
17684.079474 [NR_MAC] E current PUCCH inactive: curr_pucch frame.slot 678.9 not matching with computed frame.slot 678.19
17684.079829 [NR_MAC] W UE 55b2 expected HARQ pid 14 feedback at 678.4, but is at 678.8 instead (HARQ feedback is in the past)
17684.079838 [NR_MAC] W UE 55b2 expected HARQ pid 12 feedback at 678.4, but is at 678.8 instead (HARQ feedback is in the past)
17684.082166 [NR_MAC] W UE 55b2 expected HARQ pid 6 feedback at 678.9, but is at 678.13 instead (HARQ feedback is in the past)

Assertion (found) failed:
In nr_fill_pucch() /oai-ran/openair1/PHY/NR_TRANSPORT/pucch_rx.c:93
PUCCH list is full

Exiting execution
0-DU: thread_run start time: 02/07/26 10:03:49.000000017 UTC [500]
17684.090007 [HW] I xran_get_time_stats(): total thread time 126956787624, total time essential tasks 74359456722, num cores used 3
```



Experimental 2: GNB Error Log Dump

DU crash log CU-DU
mode with 700Mbps
Traffic with 8 CPU
allocation

```
du 18199.670794 [HW] E Detected double sync message 13.14 => 13.15
du 18199.670802 [HW] E Received Time doesn't correspond to the time we think it is (slot mismatch, received 13.15, expected 13.14)
du 18199.671273 [NR_MAC] E current PUCCH inactive: curr_pucch frame.slot 13.9 not matching with computed frame.slot 13.19
du 18199.672905 [NR_MAC] W UE dcfa expected HARQ pid 6 feedback at 13. 9, but is at 13.13 instead (HARQ feedback is in the past)
du 18199.672914 [NR_MAC] W UE dcfa expected HARQ pid 1 feedback at 13. 9, but is at 13.13 instead (HARQ feedback is in the past)
du 18199.674911 [NR_MAC] E current PUCCH inactive: curr_pucch frame.slot 13.19 not matching with computed frame.slot 14.9
du 18199.675013 [NR_MAC] W UE dcfa expected HARQ pid 11 feedback at 13.14, but is at 13.18 instead (HARQ feedback is in the past)
du 18199.675020 [NR_MAC] W UE dcfa expected HARQ pid 3 feedback at 13.14, but is at 13.18 instead (HARQ feedback is in the past)
du 18199.677376 [NR_MAC] W UE dcfa expected HARQ pid 12 feedback at 13.19, but is at 14. 3 instead (HARQ feedback is in the past)
du 18199.685709 [HW] E Detected double sync message 15.4 => 15.5
du 18199.685722 [HW] E Received Time doesn't correspond to the time we think it is (slot mismatch, received 15.5, expected 15.4)
du 18199.689098 [NR_MAC] E current PUCCH inactive: curr_pucch frame.slot 15.9 not matching with computed frame.slot 15.19
du 18199.689453 [NR_MAC] W UE dcfa expected HARQ pid 4 feedback at 15. 4, but is at 15. 8 instead (HARQ feedback is in the past)
du 18199.689461 [NR_MAC] W UE dcfa expected HARQ pid 14 feedback at 15. 4, but is at 15. 8 instead (HARQ feedback is in the past)
du 18199.691796 [NR_MAC] W UE dcfa expected HARQ pid 2 feedback at 15. 9, but is at 15.13 instead (HARQ feedback is in the past)
du 18199.691804 [NR_MAC] W UE dcfa expected HARQ pid 5 feedback at 15. 9, but is at 15.13 instead (HARQ feedback is in the past)
du 18199.695923 [NR_MAC] W Current 16.8 : future UL_tti_req's frame.slot 15. 8 does not match PUCCH 16. 8
du /oai-ran/openair1/PHY/NR_TRANSPORT/pucch_rx.c:95 nr_fill_pucch() Exiting OAI softmodem: _Assert_Exit_
du
du Assertion (found) failed!
du In nr_fill_pucch() /oai-ran/openair1/PHY/NR_TRANSPORT/pucch_rx.c:95
du PUCCH list is full
du
du Exiting execution
du 18199.695952 [HW] I xran_get_time_stats(): total thread time 220409716462, total time essential tasks 127776502542, num cores used 2
du ORAN: trx_oran_get_stats
du ORAN: trx_oran_end
du fh_rx_bbdev-21 worker thread finished on core 21 [worker id 0]
du Closing timing source thread...
```

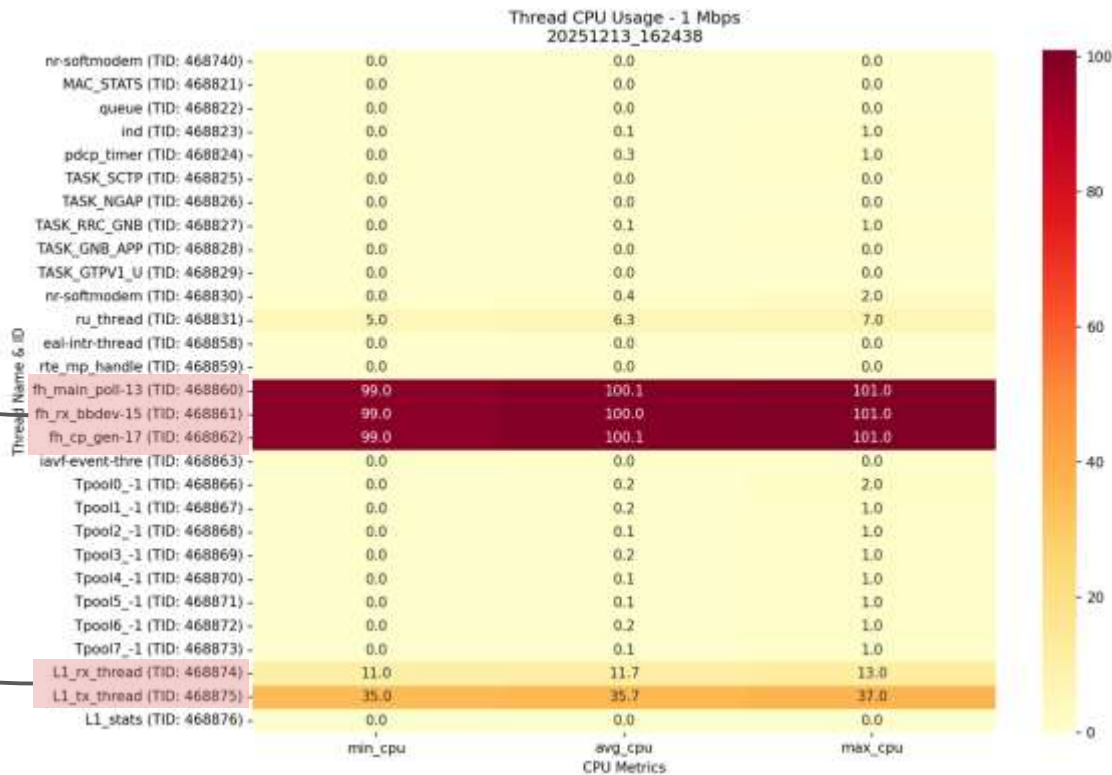



Fronthaul related thread processes.

[illegible]



Experiment 3: Sideload Profiling

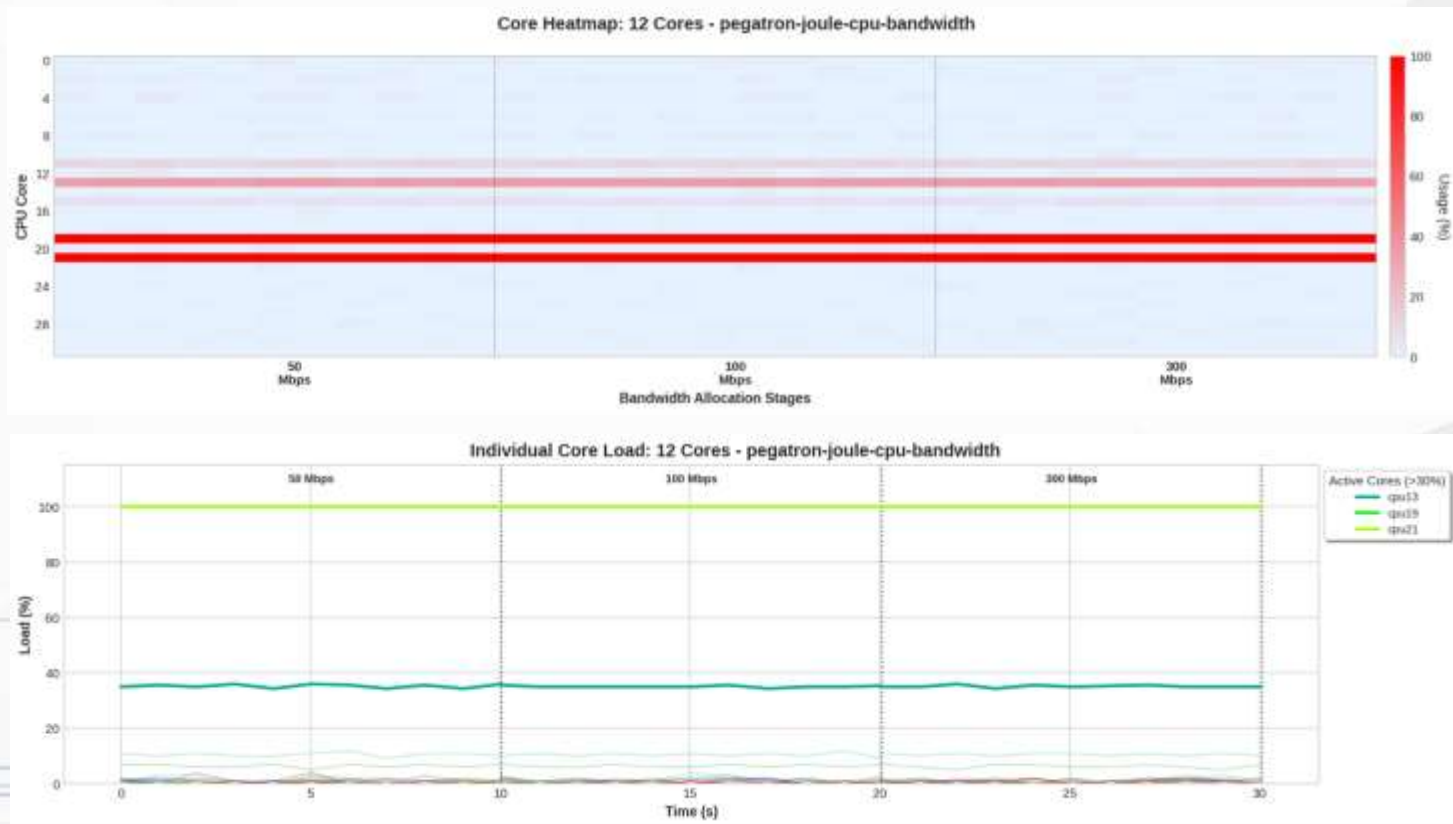


Fronthaul thread performing polling non-stop. Thread need to be clean

L1 RX and TX. On-demand. High priority process.

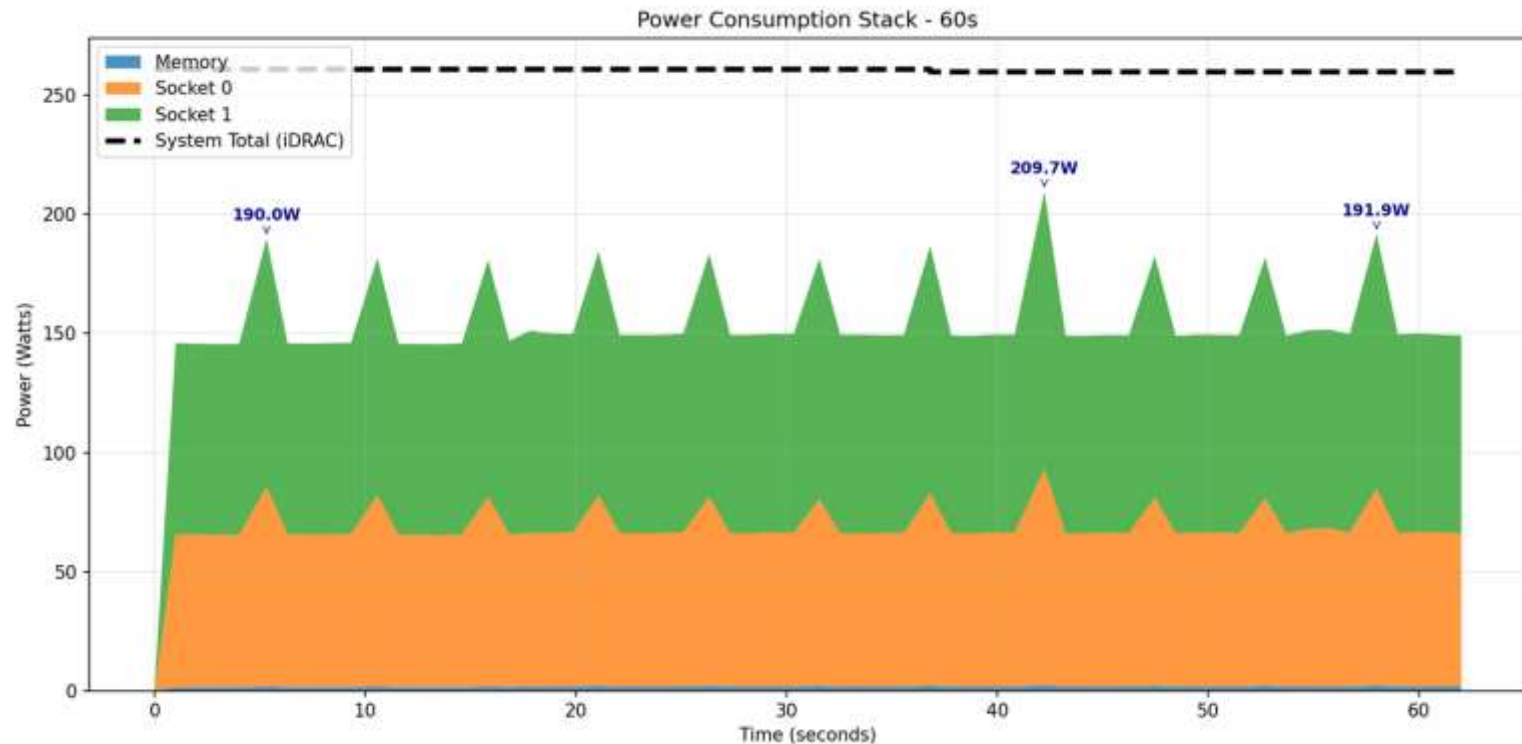


Experiment 3: Sideload Profiling Gather CPU



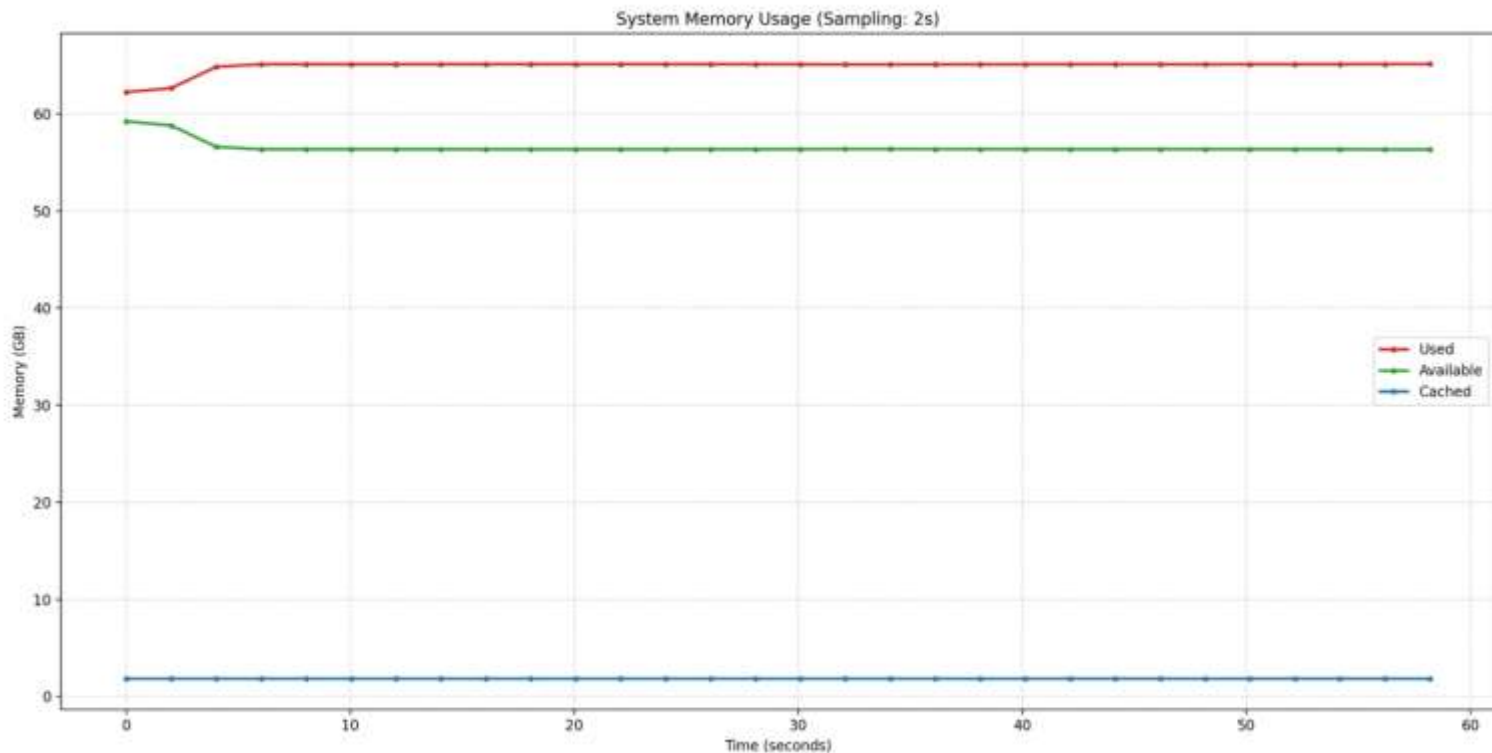


Experiment 3: Sideload Profiling Gather Power





Experiment 3: Sideload Profiling Gather Memory





Conclusion

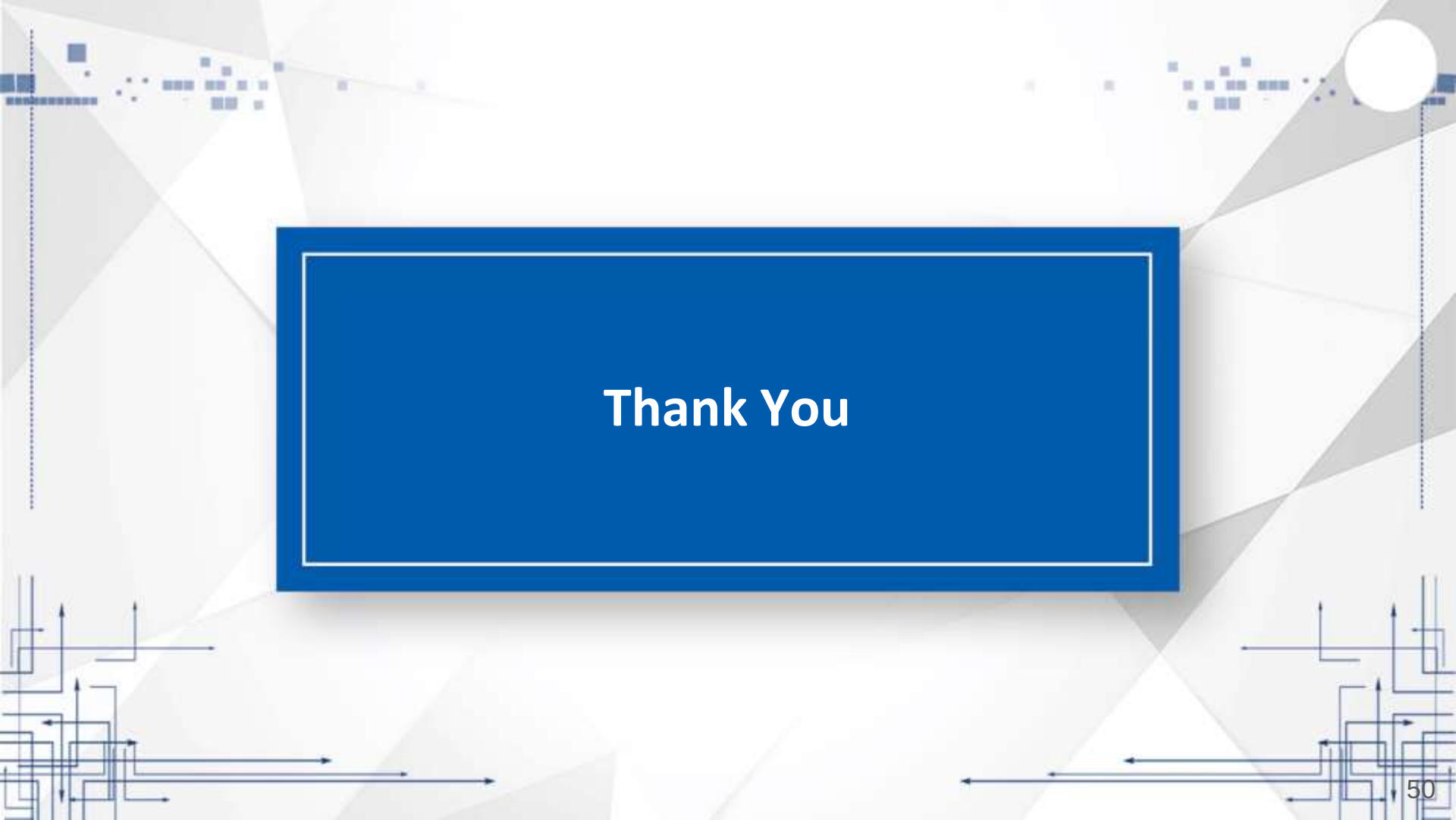
This thesis presents a platform-agnostic Automated CI/CD Testing Framework that resolves the fragmentation between heterogeneous O-Cloud environments. By implementing the O2 interface via NFO and FOCOM modules, the system decouples the management layer from the infrastructure, enabling a unified control mechanism across StarlingX and Red Hat OpenShift.

- **Unified Orchestration:** The NFO and FOCOM modules enable seamless provisioning and lifecycle management of Network Functions across different clouds, eliminating the reliance on vendor-specific tooling.
- **Automated Validation:** Integrating GitOps with an rApp-based control loop removes human intervention from regression testing by autonomously triggering deployment, validation, and teardown based on code commits.
- **Granular Visibility:** The "Sideload Profiler" architecture exposes the internal system state, capturing critical performance metrics that remain opaque in fragmented container environments across different Cloud vendors.



Future Works

- **TM500 & AIRSG Integration** - Automated multi-UE testing
- **AI-Based Auto-Configuration** - Optimize parameters based on results
- **Multi-Vendor Testing** - Additional O-Clouds (Nephio, Tanzu) and O-RUs
- **Closed-Loop Automation** - Auto-adjust and re-deploy based on failures
- **Complete O2 Implementation** - E2 and O1 interface support

The background features a complex geometric pattern of overlapping triangles and polygons in shades of gray and white. A prominent blue rectangle with a white border is centered on the slide. In the bottom right corner, there is a white circle and a series of blue lines with arrows pointing in various directions, suggesting a network or flow diagram.

Thank You



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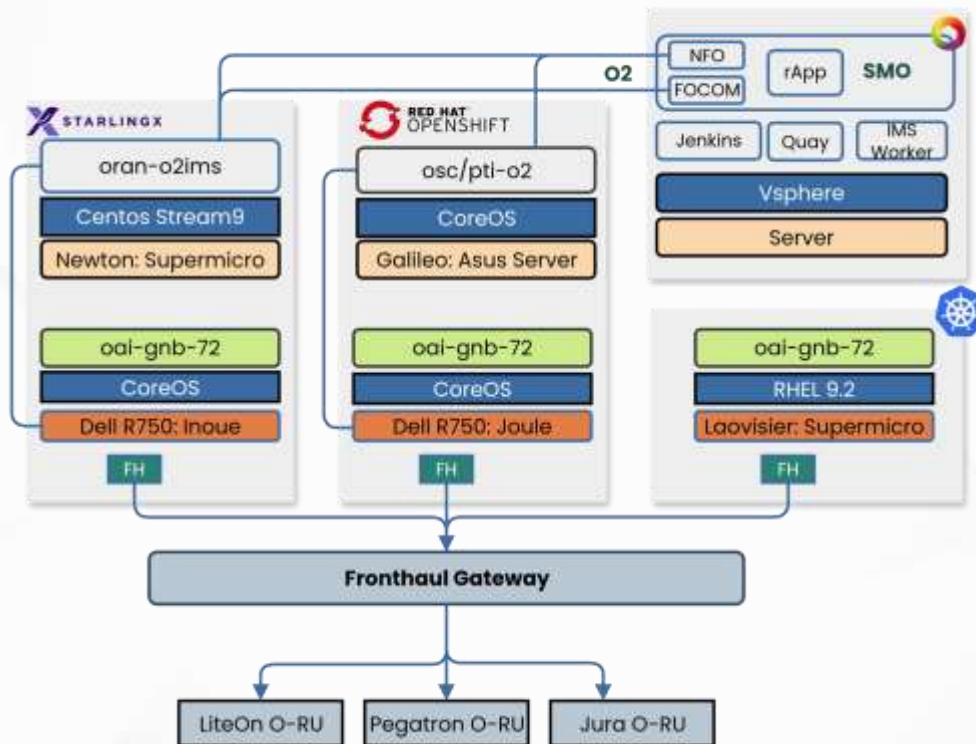


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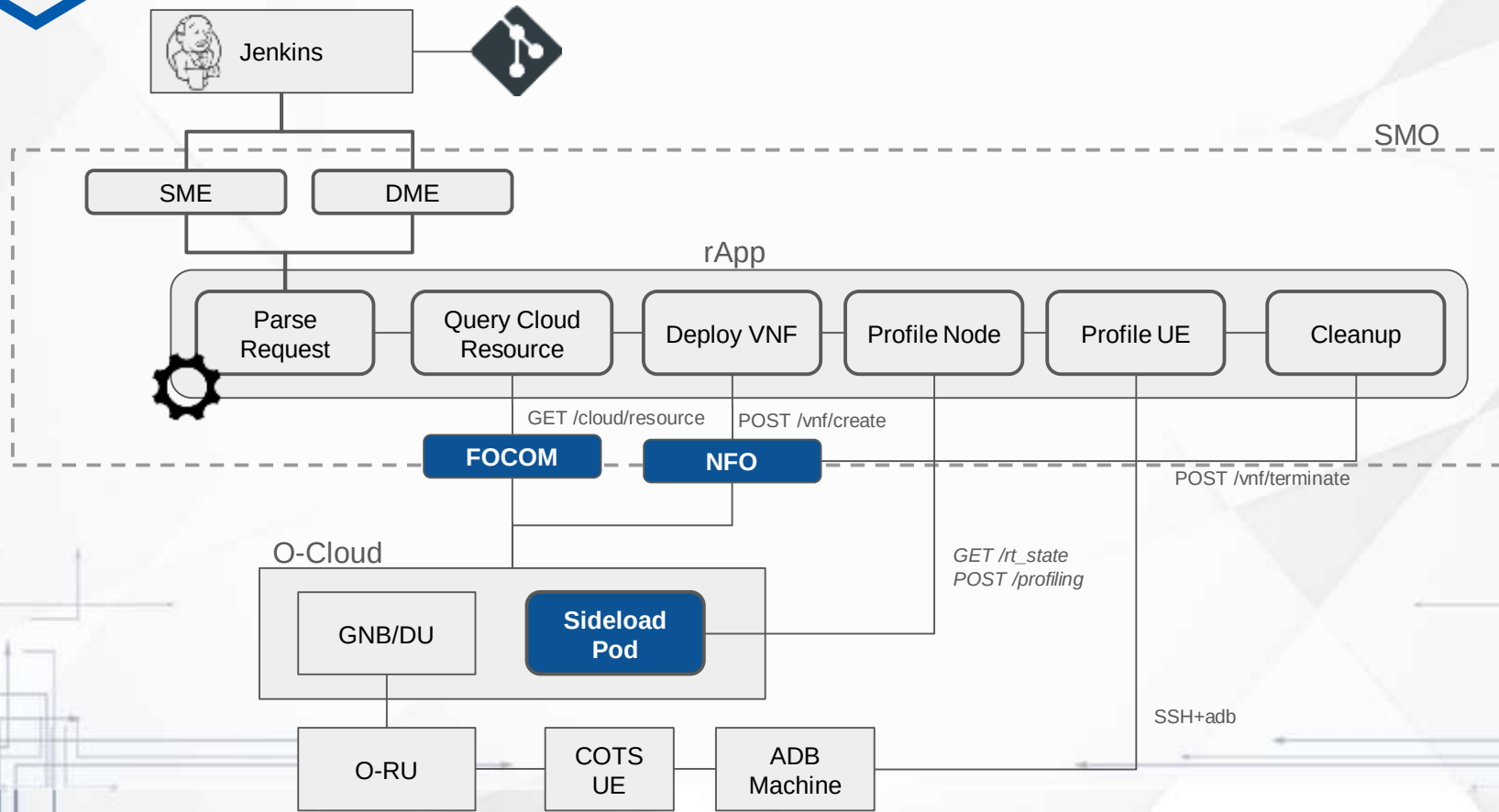
Topology

55



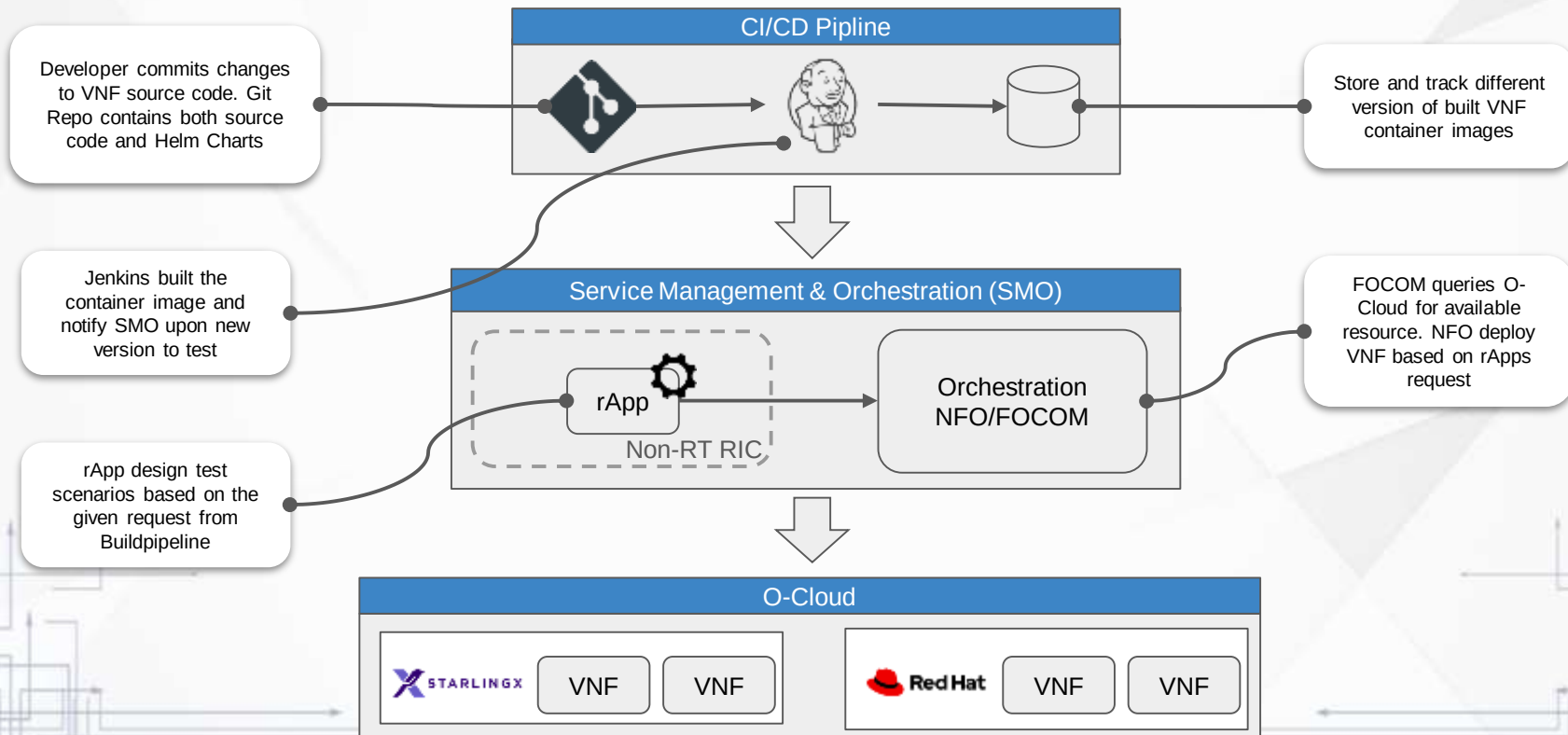


Contribution II: rApp





Proposed Method: GitOps Automated Pipeline

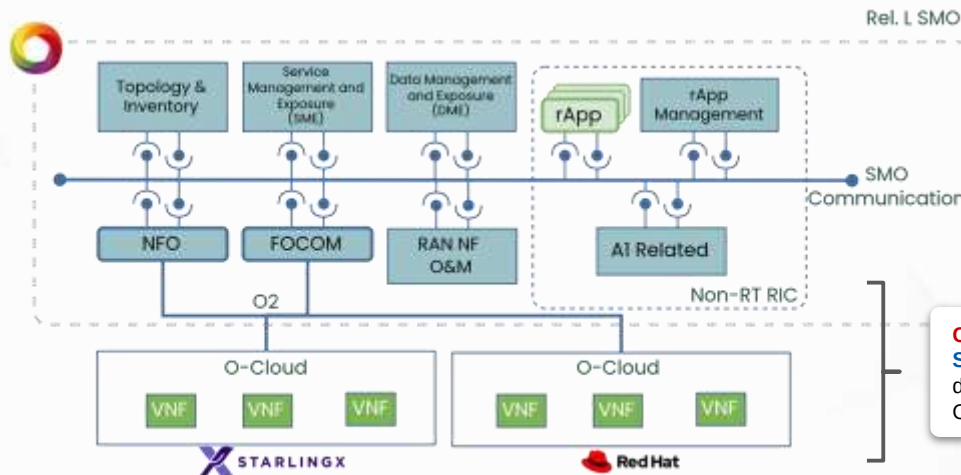




System Architecture

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CHALLENGE 2: Manual Testing with serial execution, no automation to validate RAN VNF on given O-Cloud



CHALLENGE 1: Platform-specific APIs, no portability . OSC have no implementation of NFO and FOCOM

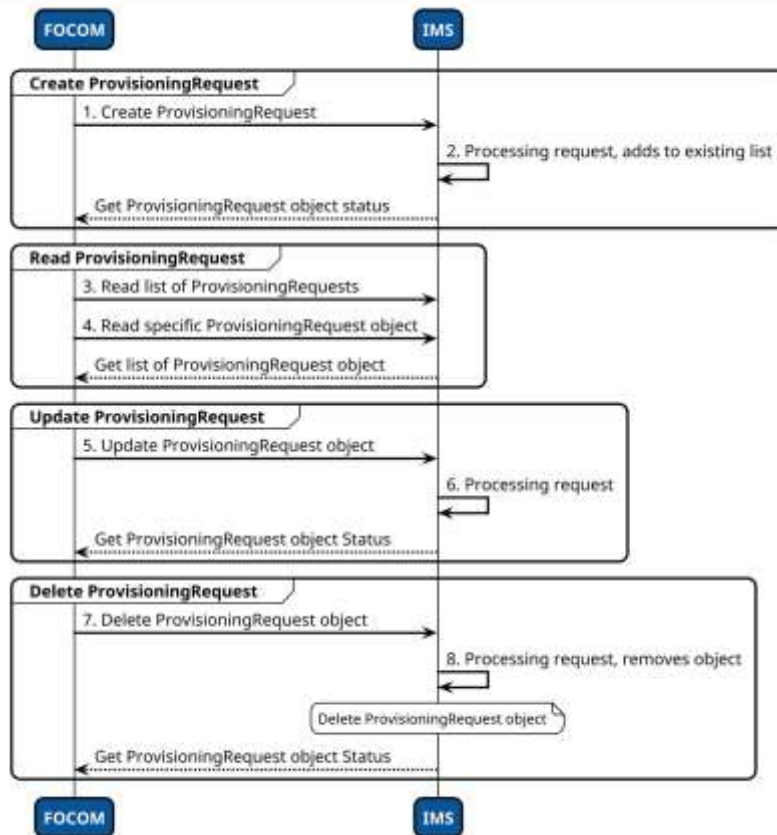
CHALLENGE 3: Lack of Standardized Tooling for different metrics across different O-Cloud vendors



O2 Infrastructure Management Services

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Goal: Provision and manage O-Cloud resources (discover, reserve)



Full Inventory: IMS catalogs all key components, including Cloud ID, resource pools and deployment managers.

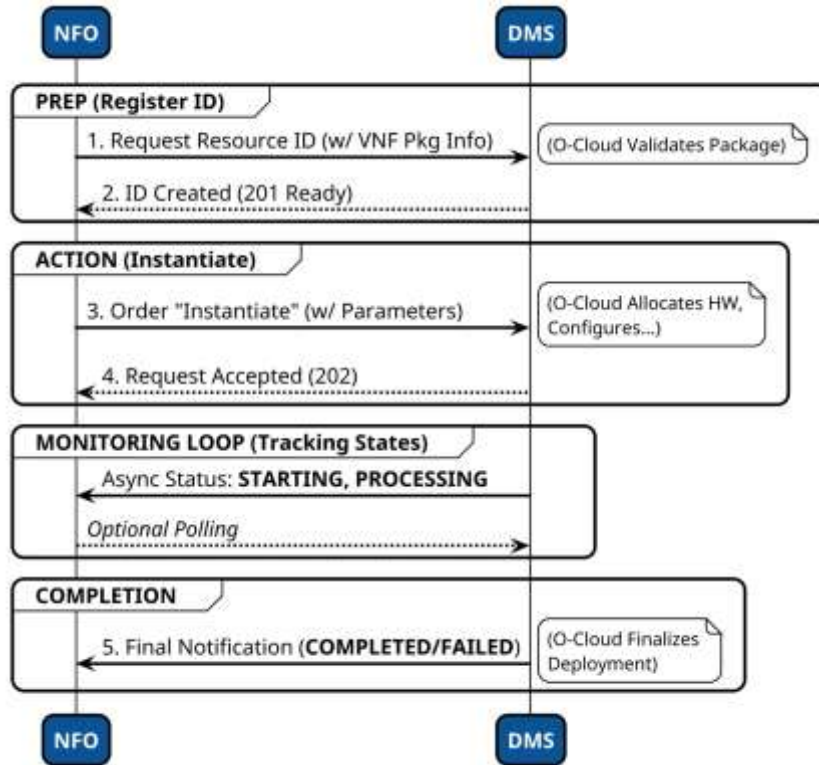
Resource Discovery: FOCOM can use IMS to identify hardware capabilities like cluster nodes, SRIOV NICs, and RT kernel



O2: Deployment Management Services

Goal: Manage lifecycle
(deploy, configure, monitor) of
Network Functions

Monitoring: Tracks reports
lifecycle states: STARTING,
PROCESSING, COMPLETED, or
FAILED



Packages: VNF packages with
identifiers (vnfpkgid) and
templates (Helm Charts) to
define the application.

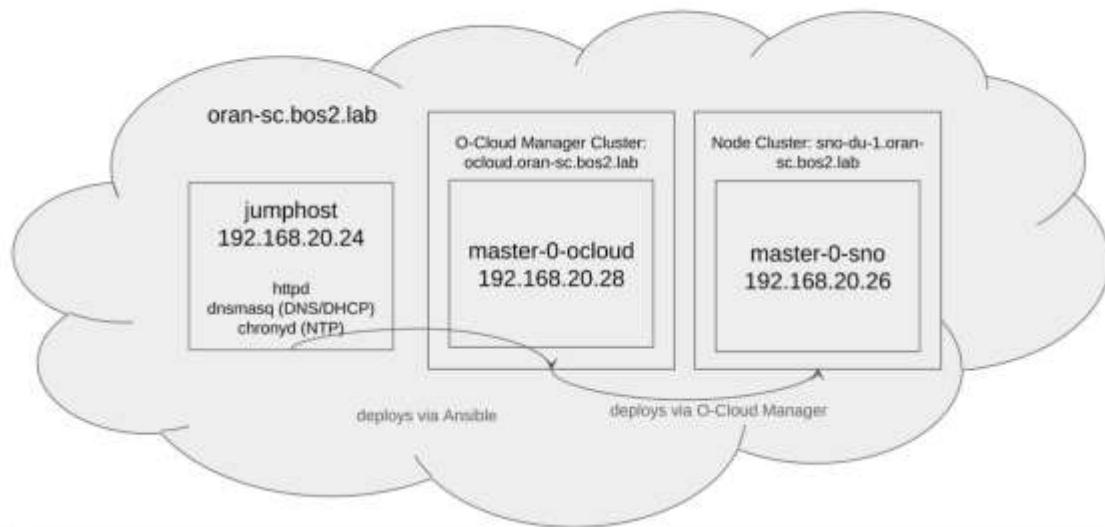
Prep: Registers resource ID using
package definition



ORAN F2F Dallas: OKD as O-Cloud [x]

- **OKD**
 - Community distribution of Red Hat OpenShift Container Platform
- **Stolostron**
 - Community distribution of Red Hat Advanced Cluster Management
 - Components:
 - **backplane-operator:** management of multiple clusters
 - **multicluster-observability:** multi-cluster monitoring
 - **cluster-group-upgrades:** lifecycle management for
 - **siteconfig:** enables cluster deployment via Assisted Installer service
- **Oran-o2ims**
 - O-Cloud Manager operator that provides O2 IMS and DMS API services

Demo Environment

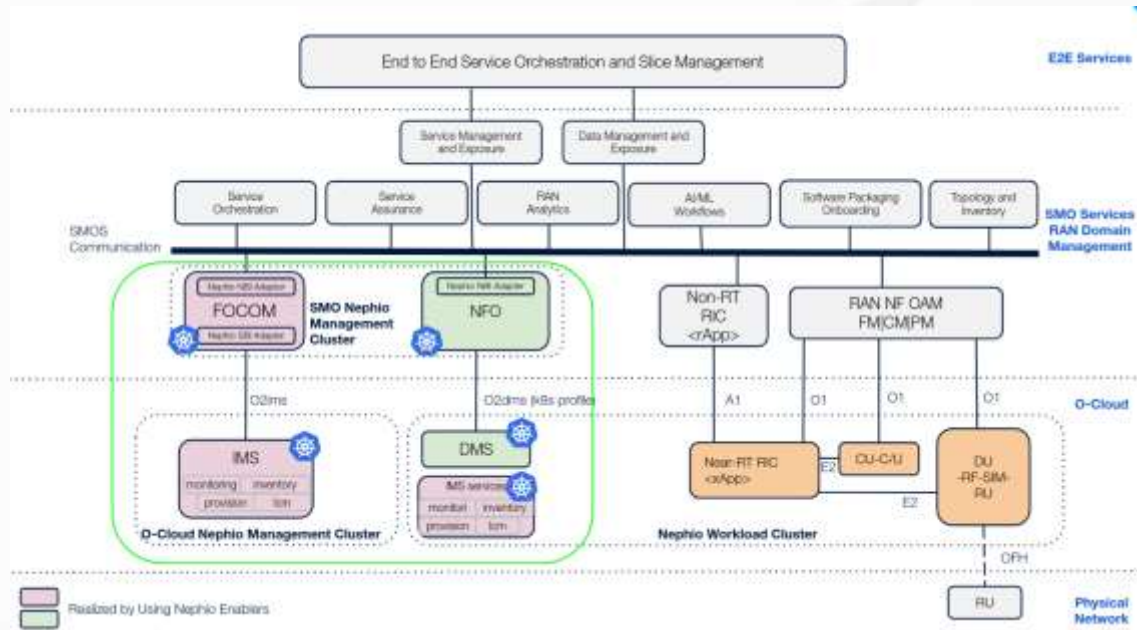




Nephio Linux Foundation Demo [\[x\]](#)

Nephio O-RAN Integration Architecture

- **Decoupled Architecture**
 - SMO (FOCOM/NFO) and O-Cloud IMS run on **separate** Nephio management clusters.
- **Standardized Interface**
 - Integration between instances occurs exclusively via **O2ims**.
- **Vendor Flexibility**
 - Ensures total decoupling between SMO and O-Cloud providers.
 - Allows mixing Nephio-based and non-Nephio implementations.

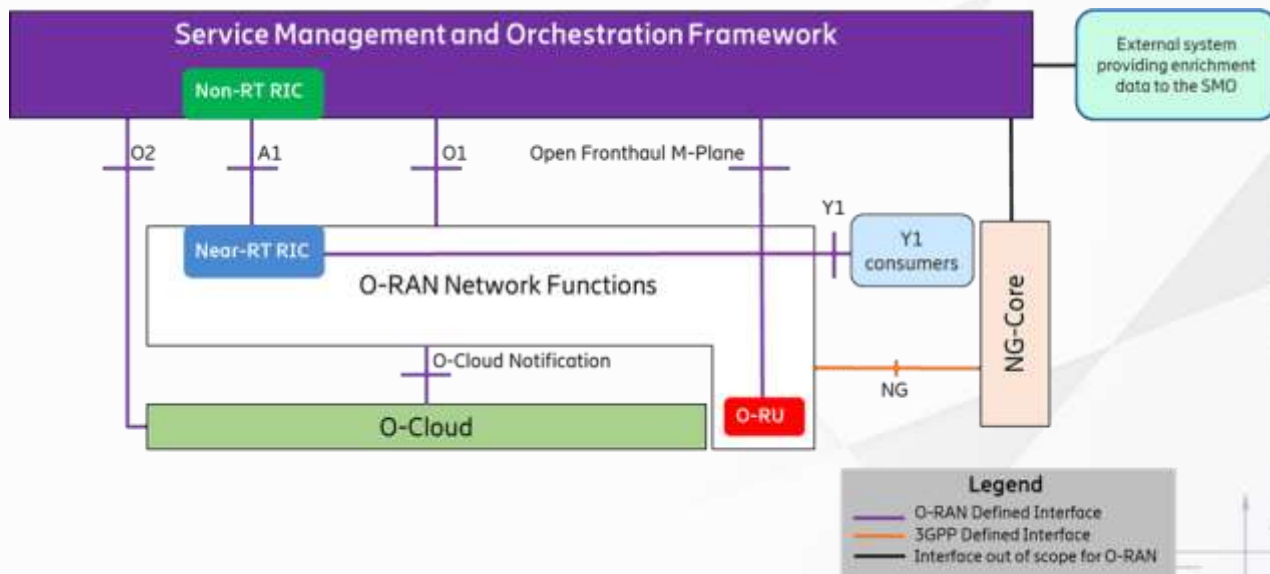




StarlingX [X]

63

- Implements the O2 protocol connecting the SMO (Management) to **O-Cloud (Infrastructure)**.
- Provides Infrastructure Management (IMS) and Deployment Management (DMS).
- Automates resource discovery, scaling, and full **FCAPS** (Fault, Configuration, Accounting, Performance, Security).
- Integrated and managed as a native StarlingX System Application.





FOCOM-Enabled Automation and Closed-Loop Control for Multi-Cloud O-Cloud Deployments [8]

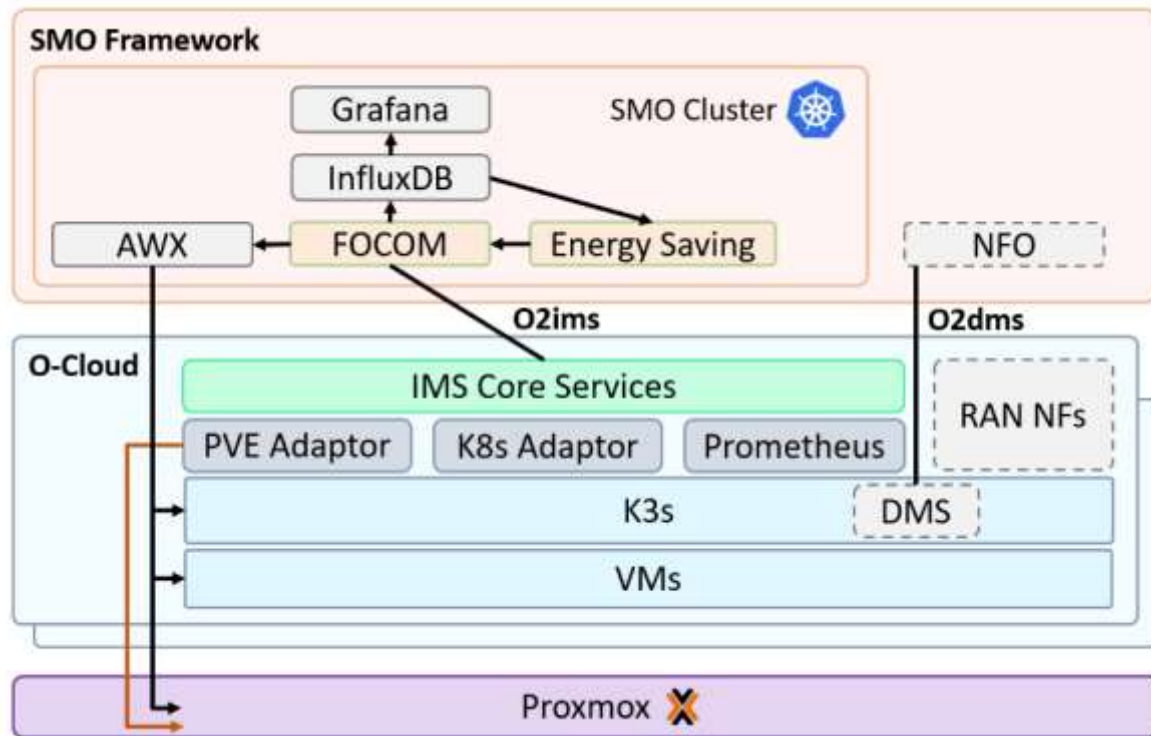
64

Contribution:

- Generic IMS Architecture
- FOCOM-enabled Automation Mechanism
- FOCOM-SMO Interworking Mechanism

Missing

- RAN Network Function workload validation
- Heterogeneous O-Cloud platform testing





5G-CT: Automated Deployment and Over-the-Air Testing [9]

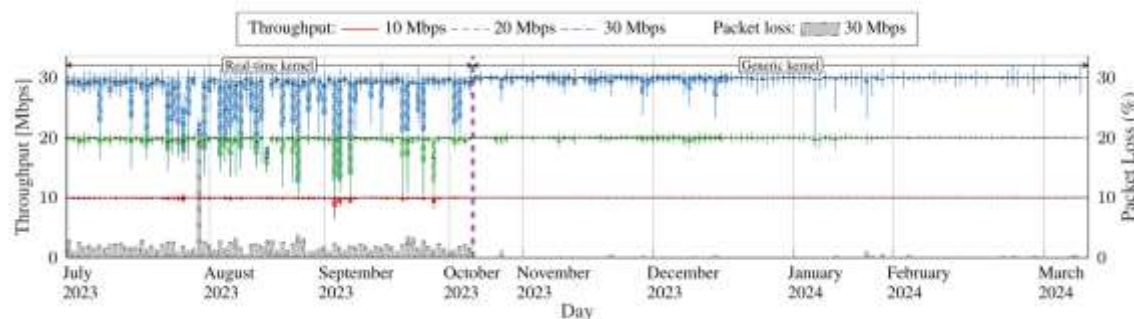
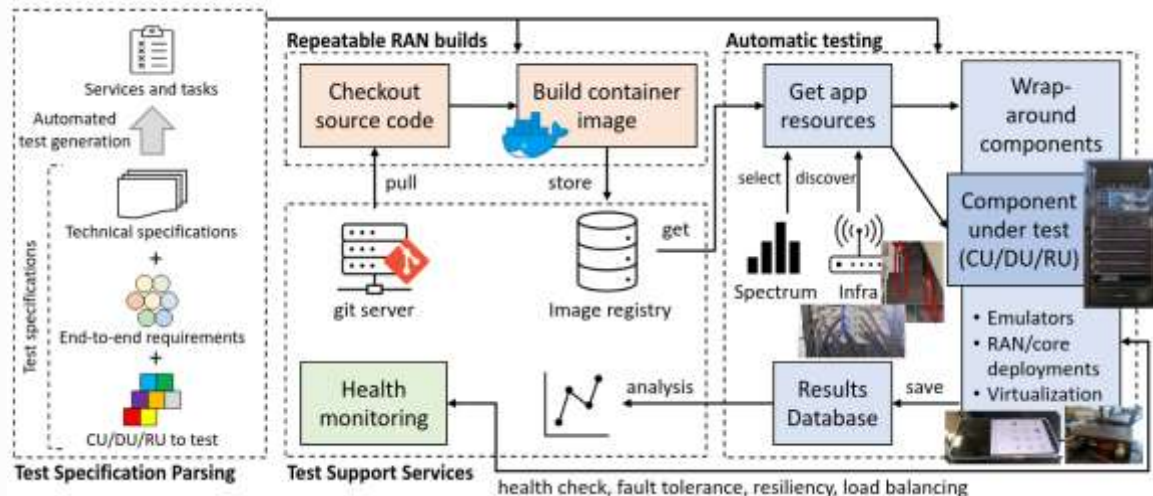
65

Contribution:

- Automated Open RAN Pipelines
- 5G-CT OpenShift based Automated Infrastructure And Microservices

Missing:

- Heterogeneous O-Cloud platform testing
- Regression test over different set config on RAN software





Blueprint for Deploying 5G O-RAN Testbeds [10]

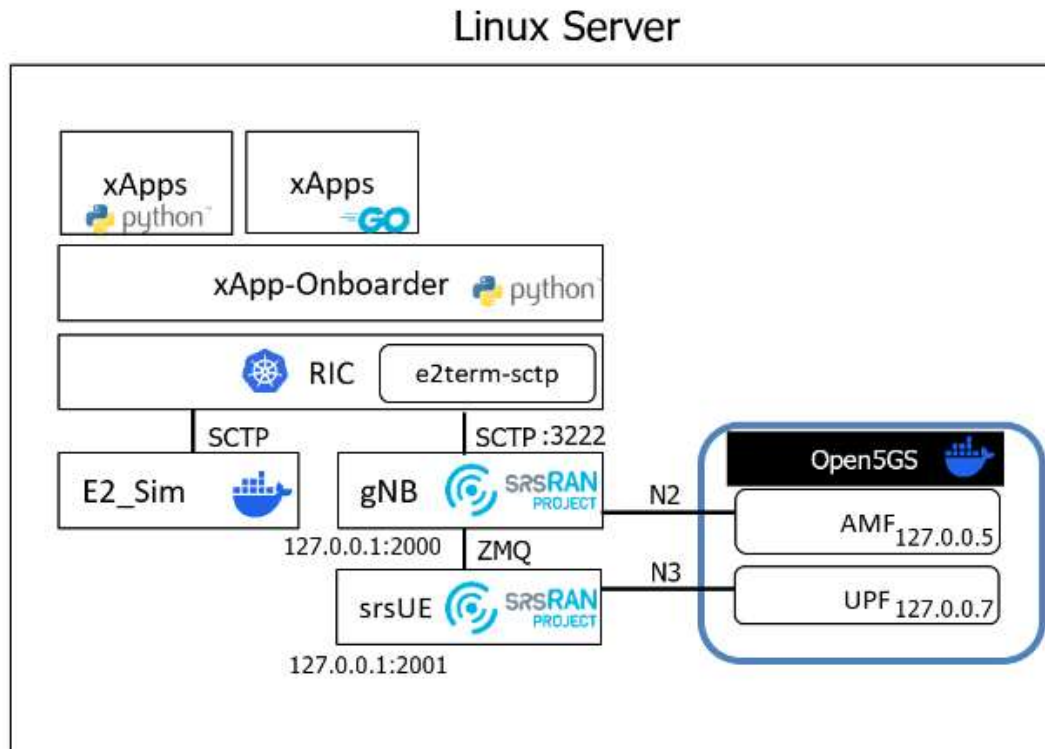
Step-by-step blueprint for O-RAN testbed deployment with multiple scenarios

Contribution:

- Covers aggregated and disaggregated deployment patterns
- Detailed configurations for ZMQ, USRP, and channel emulator setups
- Automation tool to simplify component installation

Missing:

- Automated CI/CD pipeline integration
- Integration with Physical O-RUs





FOCOM: Register IMS Stx

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```
09:14:13 root@newton ~ → curl -X POST http://192.168.8.35:5002/api/v1/ims \
-H "Content-Type: application/json" \
-d '{
  "name": "pti-o2-starlingx",
  "base_url": "https://192.168.206.202:30205",
  "ims_type": "ptio2",
  "description": "StarlingX PTI-02 IMS",
  "token_url": "http://192.168.8.69:31480/realms/master/protocol/openid-connect/token",
  "client_id": "o2-client",
  "client_secret": "u3gGUhlMkt5fyJJtUAskBoKAYTyylens"
}'
{
  "base_url": "https://192.168.206.202:30205",
  "ca_cert_path": null,
  "client_cert_path": null,
  "client_id": "o2-client",
  "client_key_path": null,
  "created_at": "2026-01-15T13:14:26.300816",
  "description": "StarlingX PTI-02 IMS",
  "has_client_secret": true,
  "id": "771fc937-2a2a-43b4-ad0e-f8067a165080",
  "ims_type": "ptio2",
  "name": "pti-o2-starlingx",
  "token_url": "http://192.168.8.69:31480/realms/master/protocol/openid-connect/token",
  "updated_at": "2026-01-15T13:14:26.300819"
}
09:14:26 root@newton ~ →
```



FOCOM: Register IMS OKD

```
09:12:23 root@newton ~ → TOKEN=$(oc create token -n oran-o2ims test-admin --duration=48h)
09:12:27 root@newton ~ → curl -X POST http://192.168.8.35:5002/api/v1/ims \
-H "Content-Type: application/json" \
-d "{
  \"name\": \"openshift-bmw\",
  \"base_url\": \"https://o2ims.apps.ocloud-vm-bmw.lab.local\",
  \"ims_type\": \"openshift\",
  \"description\": \"OpenShift KNI O2IMS\",
  \"client_secret\": \"$TOKEN\"
}"
{
  "base_url": "https://o2ims.apps.ocloud-vm-bmw.lab.local",
  "ca_cert_path": null,
  "client_cert_path": null,
  "client_id": null,
  "client_key_path": null,
  "created_at": "2026-01-15T13:12:38.905883",
  "description": "OpenShift KNI O2IMS",
  "has_client_secret": true,
  "id": "cb69bff4-2463-4377-a1b2-3a756ced5ad4",
  "ims_type": "openshift",
  "name": "openshift-bmw",
  "token_url": null,
  "updated_at": "2026-01-15T13:12:38.905886"
}
09:12:38 root@newton ~ → []
```

[illegible]



FOCOM: Query DMS OKD

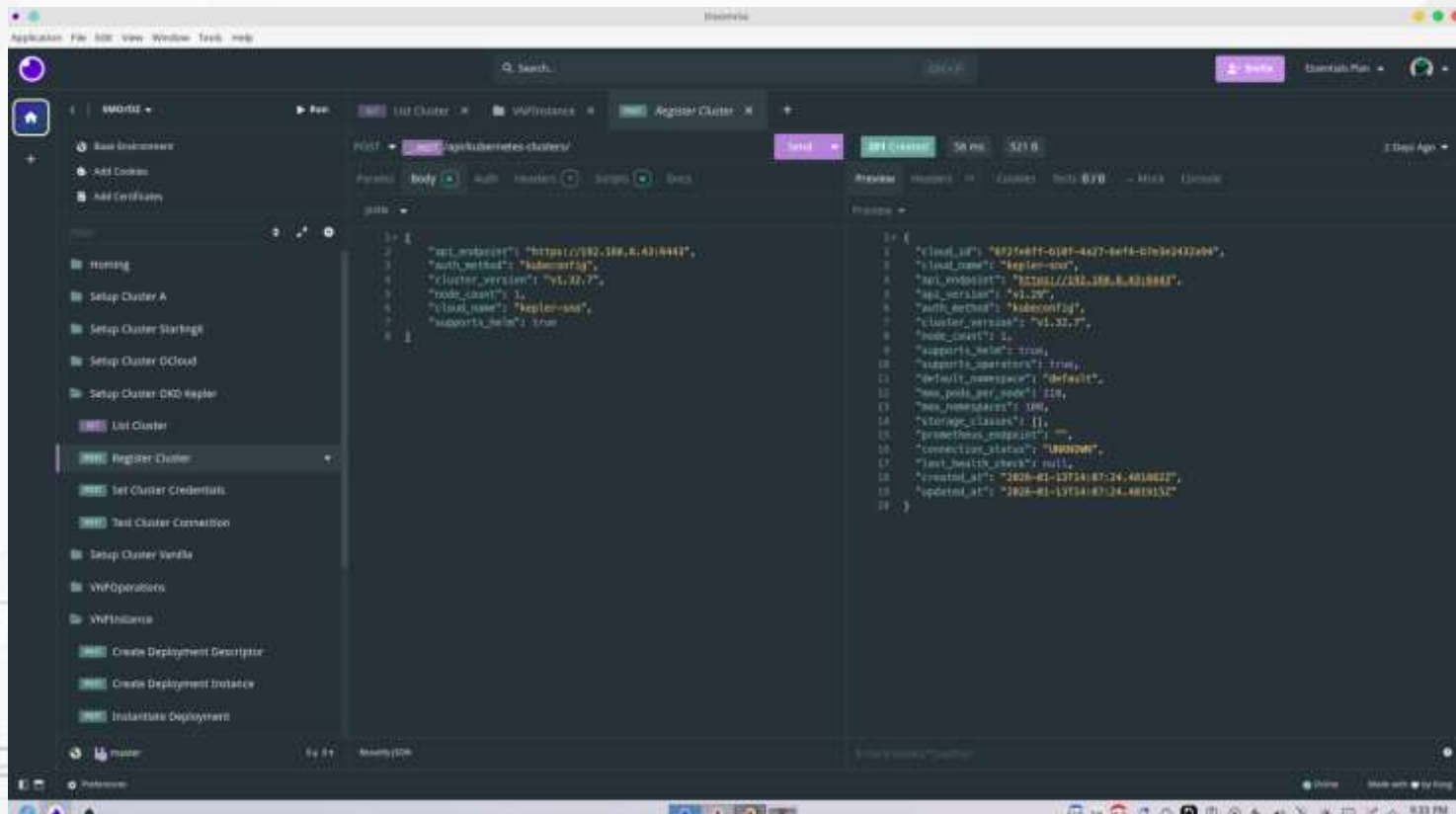
70

```
09:27:03 root@newton ~ # OCP_IMS_ID=$(curl -s $FOCOM_URL/api/v1/ims | jq -r '[] | select(.ims_type=="openshift") | .id')
curl -s $FOCOM_URL/api/v1/ims/$OCP_IMS_ID/deployment-managers | jq .
```

```
{
  "deployment_managers": [
    {
      "capabilities": {},
      "capacity": {
        "cpu": "23",
        "ephemeral-storage": "1077589412922",
        "hugepages-1Gi": "32Gi",
        "hugepages-2Mi": "0",
        "memory": "96910624Ki",
        "pods": "250"
      },
      "deploymentManagerId": "e98ad0a7-35f3-5b73-8319-3739ab421a89",
      "description": "inoue-du",
      "extensions": {
        "artifactResourceId": "79d1d40e-d322-4b16-94a9-e6e64767552b",
        "profileData": {
          "admin_client_cert": "-----BEGIN CERTIFICATE-----\nMIIDZzCCAK+gAwIBAgIIEIEMx/jxwRANBwDQYJKoZIhvcNAQELBQAwNjESMBAGA1UE\n29uZmInLXNpZ2Z5c3RlcjAe\nfW0yNjA5MDIwNDE3NTBhFw0zNTExNDE3NTFhMDAxZzAVBgNVBAAoTDRhNAwggEKA\nBAQD30Hbie7txCP23Axcg7v8ax34f+sdPKHFua3j71t6\nn97014GXgrVzMGiBSaHI+JIMq/b5EwkWKKIBYyJdVq5mfH/h3/7btqsSP5t3ZvJwp\nnT60FvXaXv+i2Z6caECsLBy1fy6Ygw5GIIo3JGppsXqHsHON9hSjv14it\n7x1p\nn81rNvb5Y6t2BRTIszDyWoRbq1iV1EjI/kJRTlafjFcU3FUw4W2Q8Pz55x70p1a1p\nnIDwn6gSp0v6rhG0rhC+rLKV7t13yN00DF34uqa1nNB9P1/D8eVpioRY8dAIM09Q\nnfn+UVJREhC8tj+KdkvyQn81y41fKe/5F\nLPE4D1mYr17AgMBAAGjZzB9MA4GA1Ud\nnDwEB/wQEAwIFoADBgNVHSUEfjAUBgggrBgEFBQcDAQYIKwYBBQUHwAwIwDAYDVR0T\nnAQH/BAIwADAADBgNVHQ4EFgQUoUoVHF5c5PC33YEAWKvrN5kiF1JGgwHwYDVR0jBBgw\nnFoAUL\nEfwFnRP150FEmxWreAu48G8IwDQYJKoZIhvcNAQELBQADggEBAIFMX+8Y\nnmUVDfM6NYxarwG98kLsm+wme8fNEA04HBWi3MspY6qWe29JHNRBy9Hg6mTpKA\nnc3Jiu9x64YIMxs13B00STKYiNnYHkpc7MUZ5MUGuIeF85\n+TAB1A+51u0jTRjan\nn+vZPcdJ/hVMudRYK7NMfyiLxZ7eXNywVwx6kZ1iLHyqjH5q8Gp2kxYKiJycwFSS\nnDVGnt35Kz+vsWVSjAA+mGt8L3RS5i1shp0Uqb3F9CYWIDtXPnaqtfb3V/7ATYCLW\nnIyHzaV8tJw/b0CuTZ\nIz3v/MiHLCV55hUSLXOR60CSW4nwLwCnW4H009a10GV\nn4r5GVNyJx42YPtc=\n\n-----END CERTIFICATE-----\n",
          "admin_client_key": "-----BEGIN RSA PRIVATE KEY-----\nMIIEPQIBAAKCAQEA99825Xu7c9j9twMXI07/Gsd+/rHrTyht7mt4+9beve0NeB1\nn4H1czBogbGhyPiZTKv2+RMJFiogwGHY1auZnx/4\n+27arEj+bd2byckU+t8b12\nn17/ottunGhArcwcpX8umIMLBIJaNyRqabNaph7BwzfYUib5eIuu+8daf/Jazb2+\nnW8rduEUYLMw81qEW6tY1dRIyP5CUU5Wn4xXFNxVM0FtkAT80kse9KdWtaSA8J\n+oE\nnqUL+q4Rtk40vqy\n744B7Tm\nn4r5GVNyJx42YPtc=\n\n-----END RSA PRIVATE KEY-----\n"
```




NFO: Cluster Mgmt (TMP Solution)





NFO: Cluster Auth

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The screenshot displays the Dissonance application interface. The left sidebar contains a navigation menu with the following items: Home, Add Cookies, Add Certificate, Homing, Setup Cluster A, Setup Cluster Starting, Setup Cluster OCloud, Setup Cluster OKD Keyer, List Cluster, Register Cluster, Set Cluster Credentials (selected), Test Cluster Connection, Setup Cluster Vanilla, VNFOperations, and VNFinstance. The main panel shows a REST client request and response for the endpoint `POST /api/kubernetes-clusters/{cluster_id}/set_credentials/`. The request body is a JSON object: `{ "auth_method": "kubeconfig", "kubeconfig_content": "..." }`. The response is a JSON object: `{ "k8s_id": "6f2f68ff-610f-4a27-b6f4-9793e2432a94", "status": "Credentials updated successfully" }`.

Dissonance

Application File Edit View Window Tools Help

Search...

Send

200 OK 35 ms 13 B

2 Days Ago

Preview Headers Cookies Tests 0/0 Mock Console

Preview

```
1 {
2   "auth_method": "kubeconfig",
3   "kubeconfig_content": "..."
4 }
```

(Show Environment)

```
1 {
2   "k8s_id": "6f2f68ff-610f-4a27-b6f4-9793e2432a94",
3   "status": "Credentials updated successfully"
4 }
```



NFO: Cluster Connection Test

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The screenshot displays the Insomnia application window. On the left, a sidebar lists various API endpoints under the 'SMO/02' collection, including 'List Cluster', 'Register Cluster', 'Set Cluster Credentials', and 'Test Cluster Connection'. The main panel shows a POST request to the endpoint `/api/kubernetes-clusters/{cluster_id}/test_connection/`. The request status is '200 OK' with a response time of '171 ms' and a body size of '82 B'. The response body is a JSON object:

```
1 {  
2   "connected": true,  
3   "status": "CONNECTED",  
4   "last_check": "2020-01-13T14:00:56.056053Z"  
5 }
```

Below the response, there is a large bug icon and a message: 'Enter a URL and send to get a response'. At the bottom, it says 'Select a body type from above to send data in the body of a request' and 'Introduction to Insomnia'.



NFO: Create Deployment Descriptor

74

```
1- |
2- | "name": "nsg-gb-test",
3- | "description": "Another VPC Descriptor",
4- | "profile_type": "hubernetes",
5- | "artifact_repo_url": "https://github.com/matanquero/ocloud-helm-templates.git",
6- | "artifact_name": "nsg-gb-fbi-k2",
7- | "artifact_repo_branch": "main/jaale-pagatree",
8- | "target_clusters": ["arn:aws:eks:us-east-1:123456789012:cluster/eks-123456789012"],
9- | "values": {
10- |   "nodeSelector": {
11- |     "kubernetes.io/hostname": "kubernetes"
12- |   },
13- |   "resources.requests.nf.cpu": 100,
14- |   "resources.requests.nf.memory": 100,
15- |   "resources.limits.nf.memory": {
16- |     "name": "nsgshift.io/fh_vpc_cp_low",
17- |     "quantity": 1
18- |   },
19- |   "resources.limits.nf.vpcplane": {
20- |     "name": "nsgshift.io/fh_vpc_cp_low",
21- |     "quantity": 1
22- |   },
23- |   "resources.requests.nf.vpcplane": {
24- |     "name": "nsgshift.io/fh_vpc_cp_low",
25- |     "quantity": 1
26- |   },
27- |   "resources.requests.nf.vpcplane": {
28- |     "name": "nsgshift.io/fh_vpc_cp_low",
29- |     "quantity": 1
30- |   },
31- |   "nfs.storage": {
32- |     "type": "fs",
33- |     "size": 100,
34- |     "storageClassName": "fh_vpc_cp_low",
35- |     "storageResourceName": "fh_vpc_cp_low"
36- |   }
37- | }
```



NFO: Create Deployment

75

The screenshot displays the NFO web interface with the following components:

- Left Sidebar:** A navigation menu with options: Home, Base Environment, Add Endpoints, Add Certificates, and a list of clusters (Homing, Setup Cluster A, Setup Cluster Starlink, Setup Cluster DCloud, Setup Cluster OKD Kedge, Setup Cluster Vanilla, VNFOperations, VNFInstance). The 'VNFInstance' section is expanded, showing 'Create Deployment Descriptor', 'Create Deployment Instance' (highlighted), 'Instantiate Deployment', 'List Deployment', 'VNFInstance Johnson', and 'VNFInstance Starlink'.
- Top Bar:** Includes a search bar, a 'Link' button, and a 'Successful Plan' indicator.
- Main Content Area:**
 - URL:** `POST /api/v2/nms/2/deployments/`
 - Method:** `POST`
 - Body:** A JSON object representing a deployment instance:

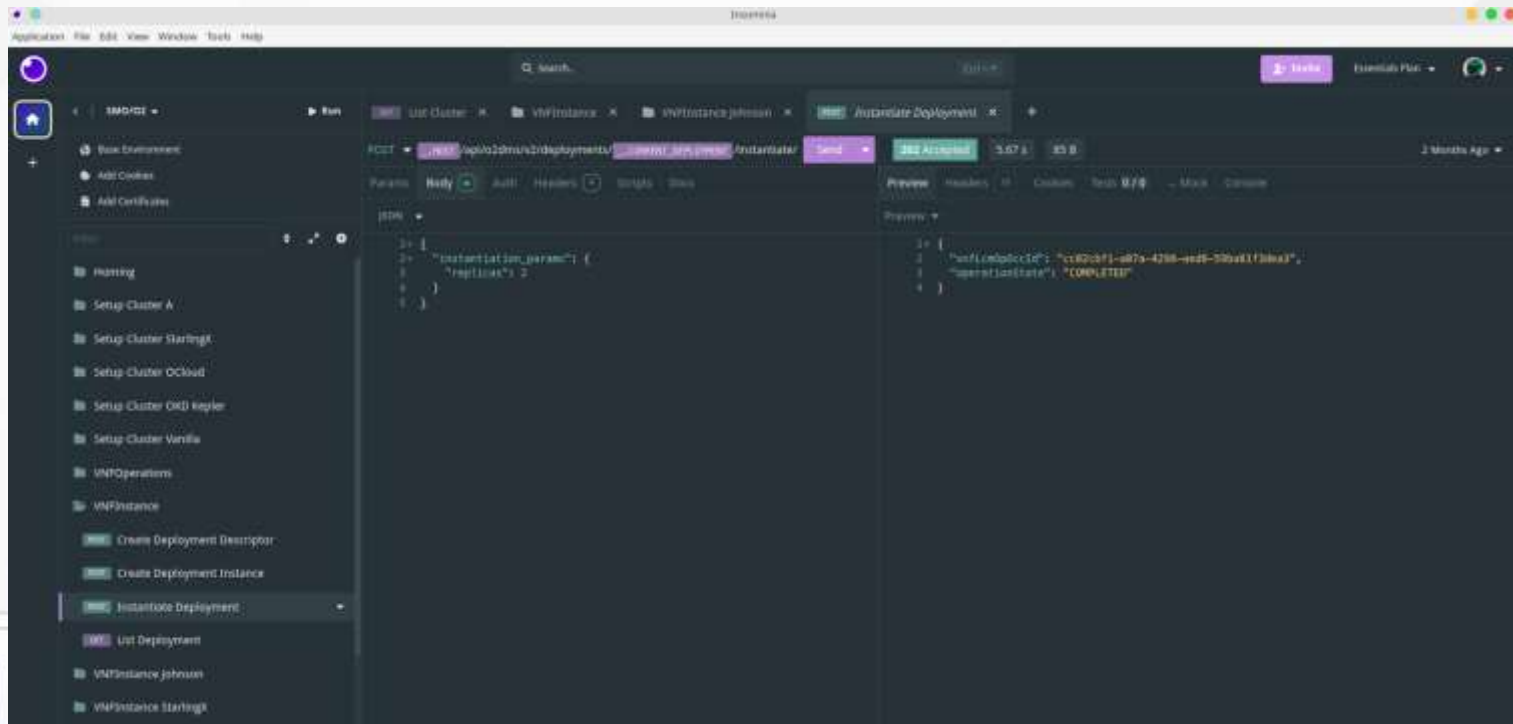
```
{  "descriptor": "test-ops-test",  "name": "test-test-abcde"}
```
 - Preview:** A JSON object showing the response from the API:

```
{  "instance_id": "4796689fb-82ef-47e1-91bb-49e339c3209f",  "descriptor_name": "test-ops-test",  "name": "test-test-abcde",  "instantiation_state": "NOT_INSTANTIATED",  "deployment_namespace": null,  "release_name": null,  "vm_instances": [],  "vm_callback_url": "",  "created_at": "2025-11-20T14:40:44.541361Z",  "updated_at": "2025-11-20T14:40:44.541361Z",  "descriptor": "C21c9254-c629-4299-ba07-85254433ab4e"}
```



NFO: Instantiate

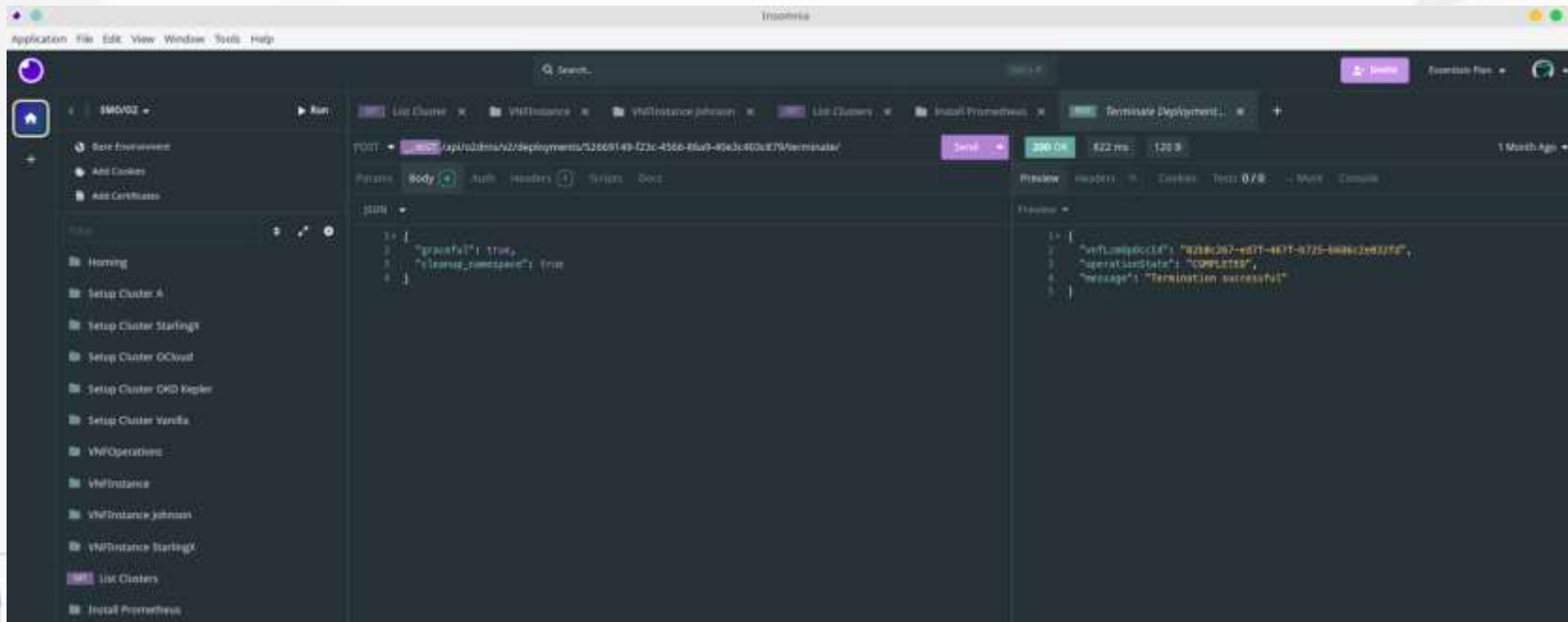
76





NFO: Terminate

77



[illegible]





Crash Log

```
889792.627006 [HW] E Received Time doesn't correspond to the time we think it is (slot mismatch, received 835.5, expected 835.4)
889792.630591 [NR_MAC] W UE 60d6 expected HARQ pid 14 feedback at 835. 4, but is at 835. 8 instead (HARQ feedback is in the past)
889792.630600 [NR_MAC] W UE 60d6 expected HARQ pid 6 feedback at 835. 4, but is at 835. 8 instead (HARQ feedback is in the past)
889792.666960 [HW] E Detected double sync message 839.4 => 839.5
889792.666972 [HW] E Received Time doesn't correspond to the time we think it is (slot mismatch, received 839.5, expected 839.4)
889792.670517 [NR_MAC] W UE 60d6 expected HARQ pid 6 feedback at 839. 4, but is at 839. 8 instead (HARQ feedback is in the past)
889792.670527 [NR_MAC] W UE 60d6 expected HARQ pid 12 feedback at 839. 4, but is at 839. 8 instead (HARQ feedback is in the past)
889792.704452 [HW] E Detected double sync message 842.19 => 843.0
889792.704464 [HW] E Received Time doesn't correspond to the time we think it is (slot mismatch, received 843.0, expected 842.19)
889792.704467 [HW] E Received Time doesn't correspond to the time we think it is (frame mismatch, 843.0 , expected 842.0)
889792.708063 [NR_MAC] W UE 60d6 expected HARQ pid 8 feedback at 842.19, but is at 843. 3 instead (HARQ feedback is in the past)
889792.708074 [NR_MAC] W UE 60d6 expected HARQ pid 9 feedback at 842.19, but is at 843. 3 instead (HARQ feedback is in the past)
889792.746957 [HW] E Detected double sync message 847.4 => 847.5
889792.746969 [HW] E Received Time doesn't correspond to the time we think it is (slot mismatch, received 847.5, expected 847.4)
889792.750550 [NR_MAC] W UE 60d6 expected HARQ pid 11 feedback at 847. 4, but is at 847. 8 instead (HARQ feedback is in the past)
889792.750561 [NR_MAC] W UE 60d6 expected HARQ pid 2 feedback at 847. 4, but is at 847. 8 instead (HARQ feedback is in the past)
889792.794475 [HW] E Detected double sync message 851.19 => 852.0
889792.794486 [HW] E Received Time doesn't correspond to the time we think it is (slot mismatch, received 852.0, expected 851.19)
889792.794489 [HW] E Received Time doesn't correspond to the time we think it is (frame mismatch, 852.0 , expected 851.0)
889792.798169 [NR_MAC] W UE 60d6 expected HARQ pid 13 feedback at 851.19, but is at 852. 3 instead (HARQ feedback is in the past)
889792.798178 [NR_MAC] W UE 60d6 expected HARQ pid 7 feedback at 851.19, but is at 852. 3 instead (HARQ feedback is in the past)
```

TYPE <CTRL-C> TO TERMINATE

```
889792.800262 [HW] I xran_get_time_stats(): total thread time 334053734538, total time essential tasks 196260682834, num cores used 3
Assertion (((prach_id >= 0) && (prach_id < 8))) failed!
In nr_fill_prach() /oai-ran/openair1/PHY/NR_TRANSPORT/nr_prach.c:75
illegal or no prach_id found!!! prach_id -1
```

Exiting execution

```
- default oai-gnb-liteon-86969f9b8b-xpv8s > gnb
```